Ep# 14 | WTF is Happening with EV? Nikhil ft. Founders of Reva, Ather, Blusmart, and Ossus

[Speaker 1] (0:00 - 4:48)

So the point of today is to talk about electric vehicles. Anybody new looking to get into the industry, we want to tell them the good, the bad, the ugly. Ready, rolling?

Hi guys, thank you for coming to Bangalore and thanks to my friends who are already in Bangalore. So the point of today is to talk about electric vehicles in the manner that anybody new looking to get into the industry after college or someone who wants to start a business. In EV, a young entrepreneur, we want to tell them the good, the bad, the ugly, which part of the industry makes sense for them to focus on and what are the skill sets they might need to inculcate to have a long career, a good career in this industry.

So maybe we can start with introductions. I'm sure many people know all of you, but let's start with Puneet. So I was born and brought up in Calcutta and I did my schooling there.

I'm born in 84, so I'm 39, shifted to Bombay in the year 2000, went to Signum College for my graduation. Where is that? Signum is in Bombay, Churchgate B Road.

Then I finished my graduation in 2005, then went to London School of Economics for my Masters in Finance, did a year of MSc in Finance there. Didn't want to come back to India and so studied one more year, went to Aston Business School for my second Masters. So while I was there studying at Aston Business School, you did two Masters in?

Yeah, in UK. One in London, one in Birmingham. Are you married?

Do you have a girlfriend? I was not focused. On girls?

Yeah. What were you focused on? Maybe just trying to find my way out, figuring out how to start a business or something.

Have things changed at 39? How is this not an interview? Things have changed a little bit.

I come from a family business background. So my third generation entrepreneurs, everybody was doing something in the family. So I didn't want to join my family business and was wanting to start on my own.

So I was trying to figure out what to do. I did two or three small startups which didn't go along well. They were in the plastic space, very small startups.

That didn't go along well. So I decided that I'll go study outside. Let's probably learn something.

And while I was studying at Aston, I chanced upon doing my dissertation. I read about the role of

clean energy and renewable energy. That was game changing for me.

And I came back to India and I met some solar panel manufacturers who at that time were manufacturing solar panels. Met them in India and realized that there could be a massive opportunity to make solar panels in India and export into Europe. Europe at that time was the biggest market.

They were setting up these massive solar power plants in Spain, in Italy, they had that fit-fit in tariffs. So I then decided that we'll set up a solar panel manufacturing facility. It was called PLG Power, my first company.

PLG, the initials of my grandfather, Pashutam Lal Goel. He passed away in April 2007 at the age of 78. So I thought, okay, my first company that I'll start should be with his initials.

And I started that in 2007, 8, January of 2008 to manufacture solar panels. The manufacturing facility was at Nashik at Cinder. So that was the whole idea to make solar panels.

So did that for four years between 2008 to 2012. The idea was to go back into solar cell manufacturing, further back into wafer manufacturing and each was a larger investment. But then the market just crashed in Europe.

It was very bad timing. Market just came down, you know, string of bankruptcies in market for solar panels. Nobody anticipated that and people were ramping up capacities for supplying panels and cells to Europe.

And we were also ramping up capacity. The market just kind of crashed in two years, between 11 to 13. And I also had not anticipated that.

And so that took a big beating. I had to shut down my first startup after achieving that, you know, that kind of manufacturing capacity. Mentally, I was also very sure.

You ran this with your family? Just myself. And big lesson learned.

I didn't have any co-founder at that time.

[Speaker 4] (4:49 - 4:50)

Just by myself.

[Speaker 1] (4:50 - 5:48)

Came away from it with some liquidity? Yeah, so I raised bank funding, raised some private equity from the market, from friends and family initial capital, then went out to raise some equity. We needed a large working capital.

We had about 150 crores of working capital at that time to buy the materials, stock it, sell it and all that. But anyway, so that the first venture shut down. Which year was this?

2012, 11-12. And so I took a big beating there. And I realized that, you know, manufacturing is tough.

It's not everybody's cup of tea to do. And I did not anticipate that this will happen overnight also. But yeah, so God shuts one door, opens another.

At that time, the government of Gujarat came out with a solar policy in Gujarat. And they allowed people to set up solar power plants. We signed a 40-megawatt power plant, which is an agreement with the government of Gujarat.

And we set up a 20-megawatt power plant. So we built that project, operated for two years, and then sold it to a large Saudi company that was looking to buy an operating asset in India.

[Speaker 5] (5:48 - 5:49)

For the sum of?

[Speaker 1] (5:50 - 6:34)

$68 million. And so that was kind of a good project. And that's when I met my co-founder, Anmol Jaggi.

Anmol is the founder of Gensol. At that time, Gensol was building many solar power plants. The solar policy was about 300.

So they were building nearly half of that capacity. Design engineering, EPC, they were doing various stuff. So I met Anmol and Anmol's company did the entire EPC.

Gensol did that 20-megawatt power plant. They built in record time of about four months. So we became friends.

And that's how I first met him. I had no idea that we'll start BlueSmart probably eight years later. So that was my second venture.

And I took a break for a year. My wife was studying in the US in Wharton. So a lot of back and forth.

Wait, you're married?

[Speaker 3] (6:35 - 6:35)

Yeah.

[Speaker 1] (6:35 - 6:38)

Why wouldn't you say that when I was badgering you with all those questions?

[Speaker 3] (6:40 - 6:44)

No, I was being nice. You did make us all feel like, yeah, you're sort of still out of the market. Really?

[Speaker 1] (6:46 - 6:54)

I'm looking at you from a different light now. Okay. For a second I was like, wait, what is going on here?

It was an absolute twist.

[Speaker 5] (6:54 - 6:55)

Okay.

[Speaker 1] (6:56 - 7:11)

Okay, then you started BlueSmart. What is BlueSmart? Tell us.

Yeah, so after my third venture, that was another small 70-odd megawatt plant. You walked away with $68 million. I didn't walk it away.

I mean, I had to give some money back to lenders also. Okay, you walked away with $50 million.

[Speaker 4] (7:11 - 7:12)

Some money.

[Speaker 1] (7:12 - 7:30)

Did you buy something nice? Some money. And then my third venture was another solar project.

Founded PLG Clean Energy Projects. We did a small 70-megawatt solar power plant in Dhulen Mastra. Again, Anmol's company, Gensol, played a part there.

So three solar ventures.

[Speaker 5] (7:30 - 7:31)

Again sold it to?

[Speaker 1] (7:32 - 13:44)

Yeah, sold it to a large listed entity in India. They're also into renewable space, a very large renewable energy company. Suzlon?

Correct, yeah. So these are three ventures and I was kind of done with solar. So you've been making money for a long time.

I lost money in my first venture. Third one? Third one, yeah.

How much did you sell that for? For $55 million. And then I kind of was thinking what to do.

And I had, I met this amazing entrepreneur called Brent Kalinikas. I'd gone to see Hyperloop One in Las Vegas and Nevada. And I met there Shervin Peshawar.

He was the CEO of Hyperloop One. And he was showing me this big tunnel where you can travel really fast in that tunnel with pods. You know, they're making these pods and you can sit in the pod.

And they travel really fast at great speeds. And they were going to do it also for logistics. Somehow in the discussion, a lot of solar conversations came out because of my background.

And the guy on the other side was Brent Kalinikas. I later learned from him over dinner that he was CFO of Uber. And Shervin, I learned later that evening, we were sitting in Malibu club and we were having dinner in Los Angeles.

And he was a series B investor. Now I'm getting a mental picture of you. Sold one company for $50 million, another for $70 million.

Sitting in Malibu club. I was learning from these mavericks what they did in their past life. I like meeting people like I'm meeting you today, I'm meeting Chetan, I'm meeting another amazing EV entrepreneur.

Okay. And then? So that evening I learned from them that what the world of mobility could be like.

So Shervin and Brent spoke a lot about EVs. In fact, when we came out of the conversation from that lunch meeting, that dinner meeting that we had, we saw a charge point, charging station, which was right outside the Malibu club. And it was charging a BMW i8 car.

And I've seen EVs before at the Stanford Shopping Center. I've driven the Tesla on around Stanford campus in 16, 17. But I was never thinking about EVs so much or the role EVs could play in the future like I thought for solar at that time.

So what is the big idea with BlueSmart? If you had to define BlueSmart in one line? So we are building an energy infrastructure and mobility company, intertwined and interconnected.

The whole idea is, which my mentor Brent also said that look, it's imminent that in the future, all cars will become EVs over the next 10-20 years. I mean, some countries will do it over 30 years, faster countries could be maybe 5-10 years, mediocre countries could take still about 15-20 years. The biggest challenge will be the charging infra challenge and that nobody's talking about.

So are you focused on the charging infra or running a fleet of electric cars? Both. Actually, it's absolutely interconnected.

If I take anything out, the entire model will crumble. And that was the whole idea. So if your company has a capex of 100 that it has done up until now, how much went towards fleet of electric cars?

How much went towards charging infra? Well, you need more money for charging infra because charging infra is a direct investment in our case, if I can speak so. And do you charge only your vehicles there or is it open for others?

Well, the whole idea is to build the breadth of charging model. I'll come there, but initially we need all the charging for ourselves. We need to help ourselves first.

But the idea, the bigger idea was to let's create an ecosystem interconnected where we will create massive EV charging infra, but that EV charging infra will be of no use if we don't have an anchor tenant. So then we need an anchor tenant, which is in-house anchor tenant, which is BlueSmart Fleet, which will then utilize the charging infra, right? And this charging infra, and then that creates a virtual cycle.

So you create charging infra, you bring in the fleet of EVs, the utilization happens. Why EVs instead of IC right now? What is the big advantage?

Yeah, I mean, EVs have massive advantages, right? So on the business side, the operating costs are super low. So the operating cost of running EVs is low.

The cost of fuel is cheaper. So look, in 2018, when we were starting up BlueSmart, the cost of CNG in India, which was a darling fuel at that time, was about 37, 38, 39 rupees a kg. And that, over the next five years, went up to about 86, 95 rupees a kg.

Now government subsidies again come down between 80, 90 rupees a kg, depending on the city you live in. And petrol and diesel at that time was about 72 rupees a litre in 2018 circa. And then that went up to now, I'm from Bombay, it's north of 105, 106 rupees a litre.

We always knew that, that this is a game of energy. And this is not a game of mobility. And if you don't build in alphabetical order an energy-infra-mobility company, I mean, it'll not make sense.

So how much does it cost you today to ply one kilometre on your EV? What vehicle do you use? I personally?

No, BlueSmart. BlueSmart, we have a fleet of 5,000 plus Tata Dekor cars. And how much does it cost you to go one kilometre in a Tata Dekor versus the equivalent model in IC?

Maybe something Ola uses or Uber uses? Look, the operating cost could range between, and that, I mean, I could speak for BlueSmart and also for an individual, could range between, you know, a rupee 20 paise per kilometre to about, maybe about 2 rupees also, depending on what your charging infrastructure is like, you know. Say for example, I'll give you a rough math.

Say I'll give you an example of MG car, you know. So, MG today, on a full charge, gives you 300 plus kilometres. And if you have a green meter installed in your house with BST in Bombay, What is a green meter?

I mean, it's a special meter that you need to take. Society will provide you a different meter, but they have a special system, a special tariff for EVs. Is that true for me?

If I have an electric car and I charge it here, I have to take a green meter? You can. I don't know about the Karnataka.

[Speaker 3] (13:44 - 13:47)

You actually haven't heard of green meters in Karnataka?

[Speaker 2] (13:48 - 14:01)

You can, but you don't have to. The rate is cheaper, right? So the rate for electric mobility may be 5 bucks.

And the rate for regular energy could be anywhere from 7-8 bucks, depending on how much energy you're using.

[Speaker 3] (14:02 - 14:08)

But for most individual owners, users, it probably will not justify the effort.

[Speaker 1] (14:09 - 14:11)

But 2 rupees a kilometre is a lot, no? No, no.

[Speaker 3] (14:12 - 14:14)

2 rupees a kilowatt hour.

[Speaker 1] (14:15 - 14:21)

And how much does that translate into per kilometre? How many kilometres do you get in a kilowatt hour of energy?

[Speaker 3] (14:21 - 14:36)

1000 kilometres in a month, you will spend 1500 rupees using a regular meter. With a green meter, maybe you'll spend 6-700 rupees, I'm guessing. He's talking of a scooter.

No, I'm talking about kilowatt hour.

[Speaker 2] (14:37 - 14:45)

A car would give you around 6 kilometres per kilowatt hour. Probably would be a good number.

[Speaker 1] (14:45 - 15:49)

The range could be 150 paise per kilometre, basically, in short. So how much is it in IC versus EV today? Today it's about 4 rupees, 4.5 rupees. The cost of CNG went up. It was only 2 rupees 75 paise. The cost of CNG went up.

Petrol, diesel today costs about 8 rupees. And IC costs you 2 rupees? No, EV costs you.

EV is depending on what your energy mix is, what your capacity is. I mean, for BlueSmart, it's a different scale, right? So our cost is very low.

Our scale is much larger. How much would it be for Ola if they had a comparison? Because they have a fleet which is...

I mean, that depends again on the fuel. If they're using CNG, the driver. I mean, Ola is an app.

They're an app. But the driver, if he's using petrol or diesel, and then depends on the car, say, Nova car or a small car. Wait, do you own your cars?

In a way, yeah. So we lease all these cars, but we have an obligation to pay for the lease. Lease from?

Leasing from DFIs. So we have secured... What are DFIs?

Development Financial Institutions, who fund these cars. Give me an example. So IREDA, Indian Renewable Energy Development Agency, So they buy it and you lease it from them?

[Speaker 4] (15:49 - 15:50)

Yeah.

[Speaker 1] (15:50 - 16:18)

So we have an SPV. They buy these cars. I mean, they fund these cars.

That SPV takes the loan from these large DFIs. And what is their contribution in the SPV and what is yours? So they give the loan to that SPV.

And that SPV, that entity, will lease cars to BlueSmart. We give the deposit, say, the margin money for procuring that loan. Can you break it down for me?

How much does this Tata car cost? Say, the car cost, say, for example, say, will be 13 lakhs. Just an average.

[Speaker 5] (16:18 - 16:19)

Okay.

[Speaker 1] (16:19 - 17:52)

And it could be lower for us at the scheme. For that car in particular? Say, for example, 12 and a half, 13 lakhs.

So we have to give certain margins. So, say, 2 lakh rupees per car as a margin that BlueSmart will provide. And then the loan is given.

And the money is paid to Tata Motors for the cost of the car. Right. And they charge you a rate of interest on that?

Yes. So there's a, yeah, so the DFI will charge the rate of interest. About?

For the loan. Sub 10%. It's about 9%.

And if anybody who, any young entrepreneur who wants to start an EV car fleet can create this entire model on their own? Yeah, well, it's just tough to do it also. Securing financing for a large scale of stuff.

I mean, that was the whole idea. But these people who are lending you the DFI, which you said, wouldn't they give them as well? Of course they will give.

They want to encourage this? Yeah, they want to absolutely encourage this. What happens to the depreciation on the asset?

If I'm a DFI and I'm lending you a 10%? Yeah. So that is where the SPV comes in.

That is where the entity comes in. So this could be exciting with depreciation. So if the entity involved could be a renewable energy company or in both businesses with also profits, you can also enjoy depreciation benefits.

40% depreciation benefits of owning and leasing that asset. So that benefits come in with the

asset owner. Who's the owner of the asset?

At the end of the cycle, these cars are valueless in a manner, right? Valueless? I wouldn't say they're valueless.

I mean, in fact, the best part about EVs is that it has life 1 and life 2. So there's life 1. But the cost of the battery, let's assume, is 60% of the cost of the vehicle.

Not really. I mean, I wouldn't say 60%.

[Speaker 3] (17:52 - 17:53)

It's really more than 35%.

[Speaker 1] (17:53 - 18:36)

It's not 60%. It could be, say, for a car, 35-40%. And when you exhaust life 1, you can obviously replace and infuse, buy a battery, work with the OEM or another startup to say, install a new battery.

That's when the beauty of EVs come in. Life 2. Because then you've already paid for the car.

The body is built to last for 8-10 years. The body is built to last for 8-10 years. The battery needs to be changed after 4 years.

So wait, if the body is built for 10 years, even if you change, how long is the battery built for? It depends on utilisation. So for an individual, On average.

Whose utilisation is only 60-100 km. On average for BlueSmart.

[Speaker 4] (18:36 - 18:37)

For BlueSmart.

[Speaker 1] (18:37 - 26:35)

I'm focusing on somebody who can start, somebody young, an entrepreneur, who can start a business such as yours. So he can go to a DFI, borrow money, lease a vehicle, run it at scale at 2 rupees a kilometre, but maybe when he begins at a higher value. Now, what is the battery life beyond which he has to change the vehicle?

And what happens to the vehicle at the end of 5 years or 10 years? Yeah. Good.

So, now for BlueSmart, our utilisation is about nearly 200 km a day. So which means in layman language about 6,000 km a month. 70,000 plus km a year.

It depends what kind of warranties or guarantees you have negotiated with the leasing institution or with the manufacturer. Some will give you say 160,000 km, some will 200, some maybe 250,000 km. So it depends how much can you exhaust.

We can utilise between 200 to 250,000 km in 4 years. Which means the battery is done. So you do 2 cycles, 4 years each, 8 years the body of the car is gone, then you have to That again depends how you use your car.

I am young, I want to start a business like BlueSmart. I am trying to figure out which DFI do I go to, how to borrow money, how to create a SPV, how to lease cars, what efficiencies do I build in to run a business of aggregating electric cars like an Uber or an Ola would do. You could start small, you may not need to go to DFI, you can go to probably small institutions who are funding cars, they are probably platform which are also leasing 100, 200, 400 cars.

Is there a cap like a DFI will only do above a certain number? I mean they, it's easier for them because the fact that they are DFI they want to give larger loans or larger amounts, it's easier for them to process. But there are small institutions also who are funding 100, 200, 300, 400 cars and I know few fleet aggregators who are also now buying EVs and deploying on the platform of Uber or trying to start micro mini fleets for B2B purposes in cities, maybe one in Bangalore, could be in West Bengal somewhere, some in Bombay also.

So we have a very different mindset. So we believe that asset financing is key for us, I mean both Moulin and myself, we have raised tons of financing for renewable energy projects so we went back to the same institution who funded us in the past and we said that you've been supporting our renewable energy story and now it's time for you to support our EV story. So is that an opportunity too?

Setting up solar power plants today? Yeah, of course. I mean Moulin continues to also run and operate Gensol, which has today actually grown to a massive company.

It builds gigawatts of solar projects today and the solar story has just begun. Massive, massive opportunity for us. What is the yield on investment if somebody were to ask?

Because I'm thinking running a solar power plant is largely commoditized by area and the amount of power you can generate. If somebody were to set up a one megawatt plant, how much will it cost and what kind of return would you make? Depends on your offtake.

I mean a megawatt plant investment could be in the range of 3 to 3.25 crores. And that number has come down from 10, 12, whatever. It used to be 18 crores a megawatt back in 2012 and came down when I was setting up my project in Gujarat was 15, 16 crores a megawatt 15.5 came down further to about 12 crores to a million dollars a megawatt back in 2017 time, 16 time. Further has come down half. And how much do you yield on that? It depends what the offtake was.

Explain offtake please. At the price at which you're selling the power. So today you need to also figure out because you have invested a certain amount you have to work out the cost at which it'll become profitable for you.

So you need to work out at the cost that you want to sell. So it determines what rate you sell power at? DISCOM does, the offtaking entity does.

What is the offtaking entity? Regulator does. So there are two ways, right?

So you can sell power to state. Like for example in my case, I sold power to Gujarat Ujjwa Vikas Nikam Limited. We entered into a fixed price contact for 25 years.

For the first 12 years the price was X, which is 15 rupees per unit kilowatt hour. And then for the balance 13 years it was 5 rupees per unit kilowatt hour, right? So it was a fixed price contract.

You could also probably do open-ended today if you negotiate with an industrial house who needs power and for power savings. Can you do that? Can you sell not to a DISCOM or the government but sell individually?

But do you have to be near the company that you're selling to? Otherwise how do you transport it? I mean, it could be near if you have empty land.

Let's say I buy, how much land do you need for 1 megawatt? 4 acres a megawatt. So how much can I sell that power for?

In India, the lowest bidding that happened at gigawatt level when Solar Energy Corporation of India, SIKI or NTPC do large bids for calling for a 500 megawatt solar investment or gigawatt investment. Those bids have been hovering around 2 rupees 75 paisa. Okay, so typically on average, what is the yield of that 4 acre land and 3 crores invested?

I mean, what is the money that you're going to make on the investment? I think about post-tax anywhere between 18 to 19 percent. Even today?

Even today, yeah. Right. And how big has BlueSmart gotten to today on the 5000 fleet?

What is the top line, bottom line? So we are nearly 440 crores around ARR. We're doing a revenue of close to 1.3 to 1.4 crores per day revenue. Good days, it's a little higher. Charging and the car put together? The ride-hailing revenue is about nearly 450 crores per ARR.

The charging revenue is about 100 crores ARR today, but that's in-house. So we have BlueSmart Mobility, which is a hold-co, and then we have BlueSmart Charge, BlueSmart Fleet and BlueSmart Tech. These are three independent subsidiaries, but all reporting to the hold-co.

BlueSmart Charge invests in the real estate and probably leases the land, builds all the charging

infrastructure, operates all the charging infrastructure. The BlueSmart Fleet, SPV, leases electric cars, pays for the drivers and runs all the paraphernalia on the ride-hailing business. BlueSmart Tech, which owns the IP, the BlueSmart app for consumers, and they all work together in tandem.

The charging company charges for the charging services which are used by the ride-hailing fleet. The fleet company's job is to bring in amounts of fleet, so we have secure financing for 20,000 cars as of now. Trying to secure financing for 40,000 cars.

And are you able to, at this scale, do it profitably? Yeah, so the charging infrastructure is profitable because we had the in-house anchor tenant, so that was profitable. Ride-hailing is a network effect game, so we have to really go deep and wide.

You really have to scale before you break in. What is the rate of growth right now? I mean, it's not massive, because it takes a lot of time to build a charging infrastructure.

So we can probably add 1,000-2,000 cars a month, but we're not able to do that. We're only able to add 400 cars to 500 cars a month. So car financing is secured.

And adding cars is easy-peasy, right? So you write a check, you get a car, the cars are ready, they're manufactured, OEMs in India can supply your cars. But again, I go back to my initial opening statement and say that this is not easy for somebody to do because adding cars is not what we want to do without availability of charging infrastructure.

The point of this entire thing is we want more young people to start working in the EV industry in one place or another. So we want to throw all the information they might require to make up their mind on whether to start a business here or get a job here. Who'd like to go next?

So Ruchi, would you?

[Speaker 4] (26:35 - 27:59)

Really? Yeah. Okay.

So this is how I thought I'd describe myself. I am a glorified brewer of green hydrogen. And let's not even give a color to it.

I'd say I'm a brewer of hydrogen. Why I say brewer is that when you're trying to break down technology so that you can pitch it to people for funding or you're talking to people to introduce them to a new molecule, you try to dumb the science down, but that's not what I want to do. My skills that I've developed over the last sort of 20 years of engineering and PhD is to be able to manipulate microorganisms, not at the genetic level, but to come together as a group and do what I ask them to do.

And I'm really grateful to them for what they do for me. Typically when we talk about microorganisms we talk about things like brewing and idli, dosa and food. That's something that's tangible.

We touch that. But my microorganisms produce electricity. So anytime we have the conversation about solar or wind, I always tell myself we should have a conversation about microorganisms that generate electricity as well.

So that's the sort of play. Still talks about what I do, but...

[Speaker 1] (27:59 - 28:00)

But give us more.

[Speaker 4] (28:00 - 29:00)

Right. Start from the beginning. I'm a child of energy.

My dad is a geologist. He started as a graduate trainee in ONGC and he was with them for 40 years. And so all I've known all my life is what is Brent price for the day?

Did we strike oil at a particular site? And the fact that my dad is, I'm really sorry dad, very active climate denier. To him everything is in cycles of centuries.

Not a single century, but multiple centuries. So he says we weren't alive and it must have been so hot so many centuries ago. We don't really know what happens and geology only gives us a clue.

The way he puts it is that geology is nothing but alcohol and good guesswork. So to him, us guzzling petrol and his very big contribution in getting all of this oil out was not the reason why we were in this situation. But I've sort of had a contrarian opinion for exactly that reason.

So I've done nothing but anything.

[Speaker 1] (29:00 - 29:41)

Just for context, would you like to explain climate change in a brief minute? Absolutely. Because everybody has a different opinion.

For a long time I assumed every surface has an albedo effect. Sun's rays hit earth. Different surfaces reflect different kind of heat back.

The atmosphere blocks different kind of rays differently by virtue of what gases are present there. The more people I talk to, sometimes somebody is telling me carbon is no longer a problem. Methane is a problem.

Water vapour is a problem. So I want to learn what exactly is climate change today.

[Speaker 4] (29:42 - 30:29)

So what is greenhouse? What's a greenhouse? You're trying to create an environment for plants where there's a constant recirculation of CO2 because that's what plants need.

Which means you're creating a dome where everything is more humid. Everything is more moist and more wet. We aren't meant to live in an environment which is supposed to be so highly heated.

I'll give you a great example and this is something that I find very troubling now. Has anybody noticed that turbulence has gotten far worse now than say a decade back? Did you know that's an effect of climate change?

Air is thinner up there than it used to be. So aircraft would hit a particular height above which there was no turbulence. And now as the air literally heats up, even at higher altitudes, you feel the effect of turbulence.

[Speaker 2] (30:29 - 30:35)

A simple way to look at it is if you're thinking of Mercury and Venus. What do you think is hotter?

[Speaker 5] (30:36 - 30:37)

Mercury?

[Speaker 2] (30:38 - 30:51)

No, Venus. And it's because of the atmosphere. So just because you're closer to a heat source doesn't mean it's hotter.

The atmosphere plays a much larger role on that front.

[Speaker 1] (30:51 - 30:55)

So climate change is the wrong kind of gases in our atmosphere.

[Speaker 2] (30:55 - 30:55)

That's correct.

[Speaker 1] (30:55 - 30:57)

Making it harder for heat to escape.

[Speaker 2] (30:57 - 31:37)

So what's happening is by the temperature going up, you're now melting the polar ice caps. You're therefore at the result of that getting changing how currents would go. So an El Nino may come in here, your monsoons may shift as a result of it because you're putting a lot more changes in ocean currents, which then change effect temperature patterns.

A rise in that would mean Mauritius may be under, Maldives may be underwater as a result of that. So it's all starting from the fact that it's methane, it's carbon dioxide as the main ingredients. There are others too.

[Speaker 1] (31:37 - 33:12)

All of what we're talking about is in a way to curtail climate change. Can each of you define climate change in one minute? In your own way?

I don't know. What I understand of climate change is obviously, to even Chetan's point is, you know, heating is obviously causing a lot of challenges. If you see the temperature which has gone up in the last 10 years, it's really bad for us.

I was reading somewhere. I think the question is why is the heating happening? It's happening because of emissions, right?

Every year, world produces 52 billion tons of CO2 emissions. Industries, more industrialization, more cars on the road, more manufacturing, more power plants. And because of that emission what is happening?

Two important contributors are, I could say, even BlueSmart plays a role there to probably curb pollution. Transportation and electricity generation. These two sectors.

Electricity accounts for 25% to 26% of the world's CO2 emissions. Logistics, delivery, mobility, all put together. Transportation as a whole, not just cars, contribute, and you can say airlines are also a part of it because airlines also contribute, again another 27%.

52%, this is Bill Gates' book which he wrote, that 50% of the global CO2 emission that happens is because of transportation on ground. So if you look at the correlation between the temperature and look at the correlation between the concentration of CO2, they have a direct correlation.

[Speaker 2] (33:13 - 33:40)

So that's very critical because the curve is like that right now. All we can do today is, it's going to continue, what do we do to reduce that kind of slope of that? So if you reduce the slope, we may hit this point 100 years later.

If we don't, we may be 20 years from today. So all we can do is probably reduce that slope. If we do it today, the impact is much bigger.

We start to do it 10 years from today, then you need to have a much more drastic change.

[Speaker 1] (33:41 - 33:42)

Tarun, climate change?

[Speaker 3] (33:43 - 34:04)

Sheeran's almost a professor. He's pretty much finished between him and Puneet. And I'm probably not the right guy to describe anything more because we didn't actually start Ather because of climate.

We're not selling you electric vehicles primarily because it will save the climate. So our view is more energy than climate.

[Speaker 1] (34:04 - 34:06)

Explain why you're selling electric vehicles?

[Speaker 3] (34:07 - 34:24)

So what we believed always is that a lot of this problem fundamentally boils down to using something that's a finite source. The electricity you're using today to run Ather is generated from finite sources. 75% is finite, but 25% is not.

[Speaker 1] (34:24 - 34:27)

Has India moved to 25% renewable?

[Speaker 3] (34:27 - 34:29)

Actually capacity I think is almost 30. No, much lower.

[Speaker 5] (34:29 - 34:30)

No, no, no.

[Speaker 1] (34:31 - 34:31)

5%.

[Speaker 5] (34:31 - 34:33)

I think either 25 or 27.

[Speaker 1] (34:33 - 34:38)

So earlier, so today, non-coal I would say, so thermal.

[Speaker 3] (34:39 - 34:39)

Non-thermal actually.

[Speaker 1] (34:39 - 34:48)

Yeah, non-thermal. So earlier that number was 95%. So renewable percent was less than 1%.

Today total installed capacity is about 175 gigawatts in India.

[Speaker 3] (34:48 - 34:49)

How much is percentage?

[Speaker 1] (34:50 - 35:28)

I would say about 26-27% is renewable capacity because the capacity utilization factor. Installed capacity of thermal is 260 gigawatts, but the CUF, the capacity utilization of PLF, plant load factor is about 70%. In solar or wind, now again wind is slightly higher on the PLF side, but the installed capacity is 175, but the capacity utilization factor, the plant load factor, who was between 20-25% in solar, now depending on new solar power plants could have higher PLF, wind is above 40.

Actual utilization of that 175 gigawatt could be in the range of about 50 gigawatt not, which is the power actually coming out, or 60 gigawatt actually coming out of the installed capacity, depending.

[Speaker 4] (35:28 - 35:30)

About 4-5% was the number.

[Speaker 1] (35:30 - 35:41)

So then you would see in layman language that it could be 1-4th of the capacity of today. So in summary, that would mean right now electricity generation 5% renewable, 95% No, no, no.

[Speaker 4] (35:41 - 35:43)

More like 25-75.

[Speaker 1] (35:43 - 35:45)

So in layman language, 1-4th.

[Speaker 4] (35:45 - 35:47)

Installed capacity basis, it's 25%.

[Speaker 1] (35:47 - 35:50)

Of the extraction also, so I said 175 gigawatt is installed.

[Speaker 4] (35:50 - 35:51)

Okay.

[Speaker 1] (35:51 - 36:18)

You are getting in terms of plant load factor, capacity utilization factor, you are getting 50 gigawatt. There, with 260 odd gigawatt as per Ministry of MNRE, I keep checking data is 270 odd gigawatts but your PLF is about 70%. What is PLF?

Plant load factor, capacity utilization factor. So for the solar plant, Just because we are speaking to like all like, you know stupid people like me who don't know much about this industry, no abbreviations, like no like simple, simple words. Correct.

[Speaker 4] (36:19 - 36:26)

So 50 gigawatt is how much, sorry to interrupt, 50 gigawatt is how much of this country's 25%, 25%.

[Speaker 3] (36:27 - 36:33)

Renewable is about 1-4th of current. Today, which is an amazing feat in the last 20 years.

[Speaker 1] (36:34 - 36:41)

I think it's In the research that I did, there are variable numbers. 5% seems to be a number from some places.

[Speaker 3] (36:41 - 36:45)

25% Yeah, it's 5% Chetan?

[Speaker 2] (36:45 - 37:01)

I think we are using it on 15%, 10-15%, not more. That's my, that's what I, you know and again the numbers vary. Karnataka is pretty high.

We are probably close to 45-45% in Karnataka. Because we have a lot of hydro and stuff. But that's not true for every state.

[Speaker 1] (37:01 - 37:13)

India's energy mix today is about 25% is renewables. That's the published data. Me too.

It's okay, we are all different places. 5%, 10-15%, 25%.

[Speaker 3] (37:13 - 37:22)

What is the loss of power in transmission in India? 10%. The rest is theft.

The 25% number that you hear is theft.

[Speaker 1] (37:23 - 39:00)

Only half of that is actually transmission. So what we are talking about is power is generated in point X, utilized in point Y. Who does this generally, like is it power grid, how many companies are there which are doing transmission?

Is there a different number where different kind of losses are happening at different transmission people? They are not South, East, West. India now has a one grid, one nation policy that came into effect in the Ministry of Power in 2003, came with the Electricity Act where they said that all these different transmission distribution companies which are now there, there should be one grid, one nation policy.

Power grid came in and said that somebody is producing power in North, East, West, South. It's one central grid of India. Everybody reports.

You can go on the website called vidyutprava.in where you can see on real time basis where the power is being generated, what is the source of that energy. In fact, that's the right website to go and check out what's India's energy mix. It's called vidyutprava.in by Ministry of Power. On average in the country would you say 10-25%? Yes, 10-12%. And that's come down post this one grid, one nation policy.

Earlier nobody was answering. Then the question to Tarun, if let's assume midway between all of you guys, let's say 15-20% of power is renewable. If you're going to lose 10-20% of that in emission, let's say we're left with 15-16% in India is renewable, 85% is fossil based.

Does it still make sense to pay the premium to buy an electric vehicle because you're net-net in today's state, putting the same amount of emissions out there?

[Speaker 3] (39:01 - 39:05)

Even if it was 100% coal based, it would still make sense.

[Speaker 2] (39:05 - 39:32)

Because? It would be around 30% better. In India's current grid, 30-35% from an ICE engine vehicle.

So an internal combustion engine vehicle has an efficiency of around 25%. An electric machine is around 90%, probably in drive cycle around 85%. So you're talking of around 3 times efficient.

Per unit of energy. So now if I take the fuel and burn it, then I'm using 3 times more energy.

[Speaker 3] (39:32 - 40:02)

The comparison of this is with the power plant, right? So the gap there is the power plant is far more efficient. The power plants are far more efficient than an ICE engine vehicle.

Scooter and cars' engines, they would be actually lucky to hit 25%. That's at the good efficiency factor. They're usually hovering in the 20 number.

But the power plants operate at like 35%. Like pretty good efficiency factors. So there all your maths work out.

The 35-10% efficiency will become 32. It's still way better.

[Speaker 1] (40:02 - 40:11)

Even if the power is generated in one way or another, at the end point, electric vehicles are a lot more efficient at using their energy.

[Speaker 2] (40:11 - 40:58)

And there's also multiple things. I think there is an issue around climate change. But there's also an issue on local emissions.

Today in cities, if you think of NOx and what's happening in Delhi, even with CNG and stuff, you know, you have zero emissions at local level. It's also easier to control emissions at single point at a power plant. It's hard to control that over a million vehicles.

And that deteriorates. So now let's look at today and 10 years from today. The ICE vehicle that's giving you X today will probably be emitting 2X 10 years from today.

The electric vehicle today, which is 30% less, will probably 70% less in the future because in the next 10 years, the grid will become cleaner. So in a way, then you start to look at the differential. It becomes far bigger.

So that's an important thing.

[Speaker 1] (40:58 - 40:59)

Zero emissions.

[Speaker 4] (41:01 - 41:18)

For me, the only concern is that imagine a scenario, and it's a best case scenario for the both of you specifically, maybe specifically for you. Imagine if all vehicles suddenly boom overnight, become EVs. Aren't you just taking the problem?

[Speaker 5] (41:18 - 41:18)

We have enough electricity.

[Speaker 4] (41:19 - 41:23)

It's not about how much electricity you have. I'm glad you think we have enough.

[Speaker 3] (41:23 - 41:52)

I just said, in scooters at least, it is 40% less emission than a petrol scooter. So all that I'm saying is that even if the grid does not become greener at all, even if the grid becomes worse, 100% coal, even in that case, you'll still be at least 30% less in emissions.

[Speaker 4] (41:52 - 41:53)

30% when it comes to...

[Speaker 3] (41:53 - 41:58)

That's a super bad scenario. It's like the world is going back. We start reducing solar.

[Speaker 4] (41:58 - 41:59)

It just lost 1.5 degrees this year.

[Speaker 3] (41:59 - 42:02)

I'm saying the world is going back on generation.

[Speaker 1] (42:02 - 42:25)

I heard the Senate hearing of Elon Musk in the US, where the senator who was questioning him said if everybody were to buy a Tesla today, you would need 20x the energy, 20x the electricity America is producing today. So are you saying the way to do this is transition slowly? 4x or 20x?

[Speaker 2] (42:25 - 42:26)

Let's look at India.

[Speaker 1] (42:26 - 42:30)

India has around a park size of 250 million vehicles.

[Speaker 2] (42:30 - 42:33)

Scooters, cars, buses, trucks, everything put together.

[Speaker 3] (42:34 - 42:35)

300.

[Speaker 2] (42:37 - 44:22)

So let's look fast forward to 2030, say 450 million. Let's say if you have some connected vehicles and more, maybe 400 or 350 to 400. Now, if everything was electric, the park size, which is highly unlikely, you'd need around 300 gigawatts of renewable energy.

India is targeting 575 by 2030. So now to put that in context, that's 0.5%, less than 0.5% of the land size of Rajasthan with solar panels. We're estimating two-wheelers maybe 50 to 70% by 2030.

Cars maybe 30%. Three-wheelers maybe 100% by 2030. Buses and stuff, that's only new vehicles.

But that new vehicles is, say, 25 million a year and then you're saying maybe 50% of that is electric. It's 10 million. But then if the park is already 300, you're adding 10 million more.

So in the next five years, you may be adding 50 million. So 50 million of that is still 25%, or less than that 20% of the park size. I've given you the example of everything going electric.

But if you then just look at that piece of it, you may need a significantly less amount of energy. Last year, India added probably the amount of energy for air conditioners that we've just added in India could power over 5 million vehicles. So to put that in perspective, we forget that what our LED TVs are doing and what air conditioning is doing, and the amount of energy we're using for that, compared to, guys, we put a million vehicles on our roads, but a lot of them was two wheelers.

This is with cars and everything.

[Speaker 4] (44:22 - 44:26)

And all of these are essentially burning coal, if not in your vehicles.

[Speaker 2] (44:27 - 44:40)

No, no, the numbers come down from 100 to 70. You're talking about once you buy the electric vehicle. Today, it's 30% better.

Next year, the energy grid gets better. It's 35. The following year, it's 40.

Then it's 50. Then it's 60.

[Speaker 4] (44:40 - 44:51)

What happens to the increase in temperature though? So you're saying the trade-off is still in favour? All you're saying is...

We're trying to still control the 1.5 degrees somewhere here, right?

[Speaker 1] (44:51 - 45:09)

I think the point they're trying to make is if the renewable mix in energy generation is going up at the pace that it is, electric vehicles will make more sense even in the future. And also what you have to see is that certain policies...

[Speaker 2] (45:09 - 47:10)

But that's changing. No, I'm saying... Let's put it...

Today, policies are changing. Right now, at 100 kilowatts, you can now get a connection. Open access, which is earlier a megawatt.

In some cities, some states, it's 500 kilowatts. Right? We're talking to them to accumulate demand, which is not happening.

But if the government now talks of accumulating demand on energy infrastructure, I could ask, Nikhil, hey guys, here's a 10-year purchase agreement, right? Put up a solar plant outside Bangalore. I don't need it here.

I'll buy energy for you 10 years. You can make your 15% return, 17% return on your money, and I will get energy 20% cheaper than the grid. And it'll be green.

My customers and environment is good, and everyone makes business. You could not say this five, seven years ago. Today, you can say that I can actually make business sense to put renewable energy in place.

Some policy frameworks have to be cleaned up. If that policy framework is cleaned up, in Delhi, we need 10 megawatts. I get it overnight.

You need probably 50 megawatts or 40 megawatts. We get it overnight. Which means all of us in the business will do that.

If he has an ether network on his charging infrastructure, and he can accumulate the demand, he says, I'm going to buy this much from you because I'm using this demand. In the next couple of years, what has to happen is energy and mobility have come together. These departments have been separate.

Once they come together, and they look at the bigger picture of 80 billion of imports, 100 billion of imports of oil, what we've talked is one is climate. The second is really the energy equation for the country. What's driving policy is energy equation and climate.

Energy equation is much bigger. What we've talked about by 2030 that all surface transportation could be electric, that's 100 billion to 150 billion dollars of less import. What does that do to the economy?

I think we have to look at it from both sides.

[Speaker 1] (47:10 - 47:36)

It was double. It was 50 billion dollars in 2018 when we were writing the white paper for BlueSmart. Now that number is 120 billion dollars.

In fact, Nirmala Sitharaman in the last budget said that first time it was 100 billion last year. It's projected this 120. So in five years, 2x.

Suruchi, why don't you finish your introduction? Can we start this conversation off with what is green hydrogen and the fuel source? Green hydrogen is creating hydrogen from a renewable source.

[Speaker 2] (47:37 - 47:52)

Gray hydrogen would be, I take methane and I split it. So I'm still putting emissions out and I get hydrogen or ammonia, depending on how you think about the structure. So green is from renewable.

I take wind energy, I take solar, I take any renewable energy and create green electrons. So that's green hydrogen.

[Speaker 1] (47:53 - 47:55)

And Suruchi, you take industrial waste and make hydrogen.

[Speaker 4] (47:55 - 47:58)

Absolutely. So for us, the problem here is exactly what.

[Speaker 1] (47:59 - 48:03)

Come from the back of your story so people have context as to how you started making hydrogen.

[Speaker 4] (48:03 - 48:15)

Sure, sure. So going back to the original story, I've always been in and around energy. Definitely thought bioenergy is critically important, maybe because I have a degree in the field.

[Speaker 5] (48:16 - 48:17)

Talk about your education.

[Speaker 4] (48:18 - 48:34)

Right. This table is so intimidating right now. I come from very humble engineering background.

I'm glad I have an education. It is insane to me that I have a PhD degree, but I think that just it was just something that happened.

[Speaker 1] (48:34 - 48:35)

What is your PhD in?

[Speaker 4] (48:35 - 49:02)

It's a degree known as bioprocess technology. I'm from, I got my PhD from Institute of Chemical Technology, which is used to be known as UDCT, University Department of Chemical Technology. It's a pre-independence institute that was created for the textile industry in Bombay.

Still sits out of Matunga. Unfortunately, because of the textile industry collapse in the 60s and the 70s, they had to look for other things to do, but they stuck to only one thing.

[Speaker 1] (49:02 - 49:06)

To a stupid person like me, how would you explain what your PhD is in? One line.

[Speaker 4] (49:07 - 49:59)

Very simple. I just engineered a few microorganisms to break down agricultural waste, basically crop residue, which continues to remain a problem five years after I finished my PhD, and converted to ethanol. So ethanol as a fuel additive.

It was at the center of energy biosciences and the key skills I developed was once again how do you create a community of microorganisms to do what you want them to do. But the problem we were solving was how do you prevent crop burning instead of coming up with ways of disposing of the crop in more meaningful ways. Suppose you could convert it to something more useful, which in our case we thought was ethanol.

The government was making a heavy push for something known as second generation cellulosic ethanol, which is basically to use non-food sources as a feedstock for making ethanol as a fuel additive.

[Speaker 1] (50:00 - 50:04)

How did you go from your PhD in this to starting to create hydrogen in Aussis, your company?

[Speaker 4] (50:04 - 51:55)

Oh my god. I had finished my PhD and I was working for my professor, my advisor's engineering company. It was an engineering procurement and construction company.

That's what chemical engineers do. Technologies out of our lab needed to be pitched to and we were heavily working with the PSUs to go out and talk about what value the technology can bring and that was me in the business development role. In between all of this, I had my professor's friend who was very wealthy but not formally educated.

He has a sort of sugar mill which runs one half of the year producing ethanol and the other half making sugar. One day his plant was shut down and he didn't quite understand because for some bizarre reason the notification was in English, not in Marathi as you would expect coming from Maharashtra. He came to my professor and he said what is going on?

He said you've been illegally dumping your wastewater, effluent as it's called. He said I have no idea. I give a dude money to sort of take it away from my site.

This man who has been collecting money from my professor's friend has been going around the block and dumping it right there because hey somebody paid me to take care of it. I took care of it. That's when my professor gave me a brief to say what do you think are the pathways as it's called in biochemistry or biology that we can use to treat this sugar industry wastewater.

That's when I happened to find this very neat biology which was not typical. These were microorganisms which are minor in community. They occur in our gut, our brain and soil and they produce electrons or electricity.

They're constantly stressed and I really related to them for that reason. These microorganisms, yes it's stressful. I would not recommend a PhD to anyone.

I would in fact say don't get an education.

[Speaker 1] (51:56 - 52:09)

I'll give you some context. In Swaruchi's company we're an investor and every time I've been on a Zoom call with her and heard something, it takes me a lot of time to go back and research and figure out what she said in the first place. Oh no, I'm so sorry.

[Speaker 5] (52:10 - 52:12)

I apologize for that.

[Speaker 4] (52:13 - 54:01)

So basically what happened was I needed funding to test this out and I said hey professor give me some money and he said no, sorry. I think you should stick to this. You give good presentations.

And honestly I thought I'd be a career academic. This was a break before I'd start off my postdoc in the US and I'm someone who's used to working with what's known as a mass spectrometer. It's this big device that breaks down proteins and tells you what's there and I was really keen on that.

And then I said wait what if I get funding to test out this technology and fortunately for me Shell had a program known as the Shell E4 program. I have zero idea about entrepreneurship at this point. I'm pretty sure I misspelled it all the time as well.

And once I joined the program they gave me some money to test it out. The only credit I'd give myself was that I had a very specific problem I had identified for them. Fracking back in 2017-18 I was producing more water than oil.

And here I had the opportunity to produce green hydrogen using these microorganisms and simultaneously recycle water. It was a win-win. But this was not a technology meant for India.

I was ready to go off to the US as I had planned. And in 2020 when did COVID happen? About four days before lockdown in Bangalore I had my tickets booked and I had to sort of shut shop and say can't do anything.

But one thing led to another. We got Tata Steel as a customer and there's not been much of a turnaround after that. Good thing we did was select green hydrogen as a molecule.

There was no policy. I'm pretty certain people when we had demo day pitches people would ask us what do you use the hydrogen for. So it's been a long journey to this point.

[Speaker 1] (54:02 - 54:06)

I still want you to explain in a very layman way what exactly you do.

[Speaker 4] (54:06 - 54:13)

We're producing electrons which is used for hydrogen gas. The water is intact and is recycled for reuse on site.

[Speaker 1] (54:13 - 54:15)

Okay, so there's dirty water comes to you.

[Speaker 4] (54:15 - 54:16)

Yes.

[Speaker 1] (54:16 - 54:17)

You create electrons.

[Speaker 4] (54:18 - 54:18)

Yes.

[Speaker 1] (54:18 - 54:19)

Where is hydrogen coming from?

[Speaker 4] (54:19 - 55:19)

Think of it as this vessel, right? There's waste water in it, not demineralized or desalinated water. Dirty water is sitting in that.

You have a couple of electrodes placed inside. On one electrode you have microorganisms. So they're trapped onto the electrode surface and they're doing their business of producing electrons.

There's a difference in potential between the two electrodes. That's why the electrons which are produced by the microorganisms are carried from the anode to the cathode which is the opposing electrode. Two electrons, two protons which are coming from water combine to make hydrogen gas.

So hydrogen gas is the product that you get out of the reactor. Electron is an intermediate but the source of the electron is biological in nature and not electrochemical. So it's not coming from solar, it's not coming from wind and the added advantage is that here's some clean water for you to reuse.

We request you to please reuse it. It's free of cost.

[Speaker 1] (55:20 - 55:28)

What is ORSYS doing now? How much have you scaled to? Who is this feasible for today to use ORSYS?

[Speaker 4] (55:29 - 57:05)

Anybody who has wastewater at their site and also requires green hydrogen is obviously our beachhead market. You're doing away with the distance problem which is honestly what people talk about when it comes to green hydrogen. You would much rather produce green hydrogen as a storage molecule during renewable energy production which is a whole other can of worms I'm about to open now.

So anytime there's surge so you have all of this power that you can't put in the grid whether it's wind or solar you can use pumped hydro and there's other sort of strategies for storing that excess power. Otherwise you're having to scale back or worse completely stop production. If you use that excess energy to split water voila you have hydrogen gas that has industrial use etc etc but it's basically used to store energy that you couldn't otherwise put onto the grid.

The problem with all of this that no one will talk to you about is that for producing one kilogram of hydrogen using by splitting water you would need depending on who you speak to nine kilograms of water which is nine liters going all the way up to 30 liters of water if it requires treatment. Treatment is basically making it desalinated water. In a country where 500 million people don't have access to drinking water India's target of 5 billion tons of indigenous green hydrogen production would require approximately 50 million tons of water.

[Speaker 3] (57:06 - 57:09)

You don't have that. Nobody supports hydrogen then.

[Speaker 4] (57:10 - 57:17)

Except if you use Oasis technology in which case you get water back again. This is a Nazi era technology.

[Speaker 1] (57:17 - 57:18)

Our company is growing really fast.

[Speaker 2] (57:18 - 57:22)

They're using it. What does it cost for a kg of hydrogen?

[Speaker 4] (57:23 - 57:41)

I can give you the good number or I can give you the bad number. Everybody wants to know. It's officially known as SHOT.

It's hydrogen priced at under a dollar. It's an Indo-US joint program. It's about 0.8 1 kg of hydrogen.

[Speaker 3] (57:42 - 57:43)

That generates how much?

[Speaker 4] (57:44 - 57:48)

33 kWh. It's the highest in energy density compared to anything.

[Speaker 3] (57:48 - 57:55)

We got 2 rupees per kW. This is really cheap. Conversion is only 30% efficient.

[Speaker 4] (57:56 - 58:00)

This is massively horrible efficiency for converting it back into electricity.

[Speaker 3] (58:01 - 58:04)

6-7 rupees. But why would you convert it back into electricity?

[Speaker 1] (58:04 - 58:05)

You would just use the hydrogen.

[Speaker 4] (58:05 - 58:07)

Yes, that's exactly why.

[Speaker 2] (58:08 - 58:10)

But you're talking for a dollar?

[Speaker 4] (58:11 - 59:18)

For certain PSUs in refining it's 40 rupees per kg. It's a very interesting chemistry. Before electrons are generated you need to break all the organic content down to what are known as short chain fatty acids.

Those of us interested in gut health have heard a lot about short chain fatty acids because it talks

about how well your stomach is doing at digesting. But it's the very same short chain fatty acids that you need as precursor. Refineries produce a lot of acids and alcohols in their wastewater which means you're starting with exactly what you need.

So that's the cheapest. Having said that, if you add compression because you need to use the hydrogen Thank you so much. It basically bumps all the way up to 112 rupees per kg.

Compared to market price, of course, it's one fourth. If you bought from certain large multinational corporations using long off-take agreements, it would cost you about 400 bucks per kg. So this is still cheaper than...

[Speaker 1] (59:19 - 59:21)

What is Oss's use case?

[Speaker 4] (59:22 - 59:27)

For us, it's completely different from the whole mobility angle.

[Speaker 1] (59:28 - 59:28)

Tell us your angle.

[Speaker 4] (59:29 - 1:00:33)

It's industrial use, right? For us, we're not even talking about steel for example. Steel is very specific.

We got lucky. We had a team that worked with us that was great. So we could accelerate very quickly.

We're looking to produce about 21 tons of green hydrogen per day at a certain site. However, going beyond that, the capacity for consumption currently doesn't exist. Even government policy asks us to wait until 2030.

Then they'll have the export system set up. Then they'll force certain industries like chemicals, ethylene, cement and steel to decarbonize using hydrogen as a molecule. But we can't really afford to wait, right?

So who are you selling to today? Those who use it as a raw material. The chemical industry uses hydrogen quite frequently in processes such as hydrogenation.

So it's not energy, it's not heat. It's basically a building block molecule to upgrade certain chemicals.

[Speaker 3] (1:00:33 - 1:00:38)

That's where the glance is, right? That's where the interest in hydrogen. They just need a lot of hydrogen.

[Speaker 4] (1:00:38 - 1:00:43)

They need a lot of hydrogen because they refine. So refineries are obviously... But the government would have...

[Speaker 3] (1:00:43 - 1:00:46)

It's not always energy. Sometimes it's a good raw material. Absolutely.

[Speaker 4] (1:00:46 - 1:02:15)

So how much are you selling now in terms of sales as a company? It's not much. It's our largest plant with another starch sorbitol manufacturer in Ahmedabad.

Our maximum capacity currently is 250 kilograms per day. And we use only 18 cubic meters or 18,000 liters of this particular effluent to do it. That's our largest sort of plant so far.

We have an offtake agreement that we're really scared of for about one ton per day for another certain client in the steel industry. But the idea is that the active working operation is 250 kilograms. It's a drop in the ocean.

For you to hit 5 million tons per year, you need about 10,000 tons per day. Fortunately for us, we know that we need only about 35 sites where there is enough density of organic matter in the wastewater. But once again, we can't really wait until 2030.

So a few months down the line, we will have to start dumping the hydrogen on the open markets. It's basically spot trading. Exactly the way LNG is traded.

And you use the same sort of LNG route. Go from Australia, Japan, South Korea, and onwards. Because there's a detailed pipeline for hydrogen that exists.

But that's grey hydrogen. The idea here is to sort of bust the market by selling it at such low values.

[Speaker 1] (1:02:16 - 1:02:32)

So Suruchi, if I were to ask you the question, not for two wheelers or cars, but for trucks or buses, is green hydrogen tomorrow going to be a viable source of energy compared to EVs, electric, all of that?

[Speaker 3] (1:02:32 - 1:02:39)

We aren't sources of energy. We are the consumers. So if they could generate hydrogen and electricity, all of it will use it.

[Speaker 4] (1:02:39 - 1:02:43)

So that we don't see as a genuine option.

[Speaker 3] (1:02:43 - 1:02:44)

Very inefficient, we won't.

[Speaker 4] (1:02:45 - 1:02:47)

Converting it back into electrons is not...

[Speaker 1] (1:02:47 - 1:02:49)

I'm talking about a bus which runs on hydrogen.

[Speaker 4] (1:02:50 - 1:02:52)

Yes, so fuel cell vehicles given...

[Speaker 1] (1:02:52 - 1:02:54)

Versus a bus which runs on electricity.

[Speaker 4] (1:02:55 - 1:03:18)

In terms of... No, please. In terms of tailpipe emissions, I think it's as good as batteries.

I think the whole equation here is that the longer the distance trucks have to go, or transportation has to go, the economics of hydrogen start to work out better. Because here are the whole transmission, storage, etc.

[Speaker 2] (1:03:18 - 1:05:10)

So you still have that ratio we talked about, right? So if you're taking the same electron from the renewable energy, you're using three times more. So whatever the cost of electricity tomorrow is, even one rupee, the cost for a hydrogen fuel cell vehicle will be three times more than a battery-operated electric vehicle.

Whatever works out on that front. But you might get a longer range. Yes, you may get a longer range.

And now with batteries, with battery swapping, you address that range piece of it in a much better

way. So where I think there could be an opportunity is in some locomotive applications where you've electrified a certain piece and you haven't done a large piece. Maybe it could be a transitional area.

That would make sense because long distance. Markets like U.S. where you would actually do a thousand miles a day and you want to run it and you run two shifts on it, right? With autonomous trucks coming in, you want to run maybe 18 hours a day.

There you're refueling maybe a thing. So it could play a role. But markets like India where you're probably doing 500 kilometers a day or 600.

You're probably stopping every three hours for a break from a driver fatigue point of view, four hours. That could fit within swapping fast charging others. So it won't make sense.

And India is a very most markets globally are very economically the commercial market is very economically oriented, right? Every market is. So unless it makes economic sense, it doesn't.

It could make sense to replace diesel. But electrics, pure electrics could still be cheaper. So it would still be a, say, if diesel is costing you X, this could be maybe 10% less and the other one could be 40%, I mean 40% less.

So in that, from that point of view, it could still be an area of zero emission and make sense.

[Speaker 3] (1:05:11 - 1:05:29)

I think hydrogen was sort of born as a concept for vehicles in a time when vehicles just couldn't do large range. Like batteries, itna bada battery lagayi nahi sakta hai. So yaar, if you can't do 500 kilometers in a truck, what's your option?

Like hydrogen is your only option. You have a different proposition?

[Speaker 4] (1:05:29 - 1:05:29)

No, no.

[Speaker 3] (1:05:29 - 1:05:32)

So that's genuine no emissions.

[Speaker 4] (1:05:32 - 1:05:42)

So the reason why the first NTPC buses started in Leh is the rarefied air and low temperature. So these are places where batteries start.

[Speaker 3] (1:05:43 - 1:06:01)

These will remain niche segments, niche use cases. But as you said, in India, now with batteries doing the range that they can and fast charging and swapping, it's becoming a very limited use case where hydrogen is your only option. And if hydrogen is not your only option, then it's the most expensive option.

[Speaker 4] (1:06:02 - 1:06:12)

I think, quite paradoxically, for some reason, it's hydrogen that's first looked at if you're talking about buses. Everybody's looking, Tata Motors.

[Speaker 3] (1:06:12 - 1:06:28)

I think it's because it's coming from that same legacy. Like it's the same set of bureaucrats, same set of consultants still writing the reports, who are still basically saying, well, batteries are $400 a kilowatt hour, you know, they can't do more than 300 kilometers of range. It's coming from that era.

Like a decade behind.

[Speaker 4] (1:06:29 - 1:06:46)

So if you look at the sort of market for clean buses, you would definitely see more tenders, whatever be the mindset for FCEVs over batteries. When it comes to buses, and I'm not even sure about distance here, how big is Leh?

[Speaker 1] (1:06:47 - 1:07:13)

Absolutely, another question. Young entrepreneur, 20-year-old going to college. Is there an opportunity to work in your industry?

And if so, what? Puneet gave us two things. He said you can set up a solar plant, earn between 15-19% interest.

Or you can raise funding, lease vehicles, set up something like BlueSmart. In your industry, what is the opportunity for an entrepreneur?

[Speaker 4] (1:07:13 - 1:09:52)

I think ours is more of an opportunity with roots in climate change as a problem. We're looking for kids who are driven by the need to change the way things are or to build a future for themselves. So they are willing to address very large problems which aren't...

So you could work for an IT services company, and I'm sure you're solving a problem which is large on scale. But this is an existential crisis. So all of that engineering skillset which would otherwise have gone into MNCs need to come into climate change and green jobs.

So I would not say it's still a nascent industry. It isn't even really an industry. It's pretty much a cottage industry.

I remember when it was just us and another company for a long time. Now you have mom and pop establishments all talking about electrolyzers. So from a business point of view, sure, if you could export, import electrolyzers into the country, you have two ministries.

So there's Ministry of New and Renewable Energy and Ministry of Power, each with different parts of the hydrogen equation that they're dealing with. So MNRE typically deals with what happens, how do you push consumption, what's known as CITE. So it's just Green Hydrogen Consumption, which it's trying to push for.

And Ministry of Power basically says, I will take anything, electrolyzers and more, as a source for green hydrogen, and I will give you whatever power you require to produce or to reach 5 million tons per day. So if somebody who is a 20 year old said the opportunity here is for me to access these bids and subsidies by getting electrolyzers in, sure, there's a business there as well. I don't think there's even 10-20% that you can make that you would probably be able to make in solar, because you'd have to spend a lot.

But, according to me, the opportunity is in trying to create massive disruption. And there's a lot of people working on making even water splitting more efficient. Sea water splitting, for example, I've heard so much about it.

So you need vocational training, more than an engineering degree right now. And if you could join the 20 companies that exist that help with this, you'd be doing your bit. So I'm not sure.

Super. It really answers your question. Very interesting, no?

[Speaker 1] (1:09:53 - 1:10:42)

Should we go to Chetan next? Chetan, would you like to? So I've known Chetan a while.

Each time I meet him at a party or somebody else's house, I keep bugging him and asking him questions. It's become like second nature. But today I get to bug you in isolation.

Get you stuck in one location for many hours. So tell us a bit about your long, illustrious journey in EVs. By the way, I have to add this thing.

I called Anand Mahindra for like, I was just trying to do research on this industry. And then I told him, you're coming. And he was like, Chetan is like the godfather of our industry.

He started this off before EVs were a thing in India. When did you begin first?

[Speaker 2] (1:10:43 - 1:10:58)

So I got into electrics in 1990. And I know, maybe I don't know if you were born at that point. No, so I got into racing solar cars when I was at college.

[Speaker 5] (1:10:59 - 1:10:59)

Wow.

[Speaker 2] (1:10:59 - 1:11:21)

And we was part of the team at Michigan. We raced from Florida to Michigan and we came first in the US. And GM sponsored us to race in Australia for the World Championship.

So a bunch of 20 year olds, we raised a million bucks, put a solar car together and we raced from Darwin.

[Speaker 1] (1:11:21 - 1:11:22)

You raised 20 million bucks? One million.

[Speaker 2] (1:11:23 - 1:11:51)

We raced, 20 year olds but you know, we raced from Darwin to Adelaide, which is 3200 kilometers in the World Championship. We came third in the world. Honda came in second.

And for me that you could travel a continent on sun energy was like, wow, this is the future. So that's what I want to do. So It had a fantastic name.

Sorry?

[Speaker 4] (1:11:52 - 1:11:54)

It had a fantastic name. I think it's called Sunrunner.

[Speaker 2] (1:11:54 - 1:12:11)

Sunrunner. So we're and then a bunch of us said that when we graduate let's start something in electric vehicles. So five of us used to meet and every two weeks and decide what business plans would come out.

[Speaker 5] (1:12:11 - 1:12:11)

The US?

[Speaker 2] (1:12:12 - 1:13:44)

The US. And then one of my friend's father who was a PhD from Caltech he invented the airbags and did a lot of seatbelt work and sold that company to TRW. He loved our idea and he was a

mentor.

He says come and work for me. So we all ended up working in California in 91 doing electric vehicles. And I went to grad school after that at Stanford and I did a lot of work on hybrid electric vehicles.

I felt that electrics needed a host of electronic software, mechanical and that was a great program that kind of combined all of this stuff. And went back to this company. It had already gone public by then.

And started doing electric vehicles with them in California. And that's the time California was switching from the aerospace industry was dying and so they were switching it to EVs. So there was a big push from both Bush administration and the Clinton administration to get electrics on that front.

And that's when the time when I came back to work for them I said listen I think the market my mentor asked on the market I think it's going to be on in India and China. And in developing countries. And the solutions we need for these are different.

So we liked the idea and the concept of REWA was born as what I would initially call an Asian electric vehicle for developing countries. And post that.

[Speaker 1] (1:13:45 - 1:13:46)

When did REWA launch? Which year?

[Speaker 2] (1:13:47 - 1:14:03)

We launched in 2001. 22 years ago. We developed in the late 90s right.

So it's it was a very different time then right because you just didn't you didn't have people you didn't have investors you didn't have the tech side of it.

[Speaker 1] (1:14:04 - 1:14:14)

REWA started in 2001. Yeah so Why REWA the name? My mother's name.

No REWA. Means a new beginning in Sanskrit.

[Speaker 2] (1:14:16 - 1:15:03)

I always thought REWA was like revolution Once you get the marketing guys it's a revolution electric vehicle alternate the EV joined together which almost every company uses that same EV that we put 21 years ago. I look at them and go I thought we trademarked that but it seems like everyone's using it. I was doing work in India and the US at that point.

Half the tech was here half the tech was there. So honestly I'm technologist I wasn't an entrepreneur right. To me I was passionate about my tech side.

But then I said okay to build a business I had to become an entrepreneur. I had a friend teach me how to put spreadsheets to create a business plan. And try to go and start to raise capital and got some seed capital for my family business.

[Speaker 1] (1:15:05 - 1:15:06)

What was your family business?

[Speaker 2] (1:15:07 - 1:16:10)

Automotive components, aerospace electric material handling. So if you look at the golf carts and most of the airports are from us. So we've been doing all of that for a while.

But this was quite different from that point of view. And so I got some seed capital there I got ICICI and technology development board which is hard because you're 29 and you ask someone that you want to get money for electric vehicles in India. The first thing they ask where does this work in the world and to give a 29 year old that was, I must say it was very good for those financial institutions to give you debt at that time because we see money and that was just non-existent at that point.

Built that out and then in 2001 we launched one month after we launched the government pulled the plug on subsidies and actually one month before we launched and double the taxation on electric vehicles Why was that? And reduce taxes on petrol vehicles.

[Speaker 5] (1:16:11 - 1:16:11)

Why?

[Speaker 2] (1:16:11 - 1:16:28)

Well look at who were the players at that time in the country and it just, you know, it didn't feel, I mean people didn't want electrics to work right and Seriously, I always thought that was just bureaucracy not getting it No.

[Speaker 3] (1:16:28 - 1:16:30)

They were already anti-electric at that point?

[Speaker 2] (1:16:30 - 1:16:39)

Well think about it, one month before all of this done, all of that disappeared. Subsidies that were there, you built the business on the fact that this was given and confirmed.

[Speaker 3] (1:16:39 - 1:16:44)

So who got the subsidy? If you guys weren't live in the market, who really got the subsidy? Nobody got it.

[Speaker 2] (1:16:44 - 1:16:46)

They just killed the whole subsidy.

[Speaker 3] (1:16:46 - 1:16:47)

Without giving it to anybody?

[Speaker 2] (1:16:47 - 1:16:55)

They didn't give it to anyone and they also increased taxes. This 2000? 2001.

Right?

[Speaker 3] (1:16:56 - 1:16:57)

Global financial crisis.

[Speaker 2] (1:16:57 - 1:21:22)

It was a big issue and so I think the only way to survive was actually looking at global markets. So we actually went to 24 countries. UK was my largest market.

So in 2005-2006 Tesla raised money from Draper. Others did, you know, Draper Fish Investor and us amongst others. So there was a bit of cleantech capital coming out in 2005-2006.

The first signs of cleantech, right? If you remember that the VC industry was changing a bit. So we got some VC money and then in 2010 with the Frankfurt Motor Show, we were showcasing our new products.

Only Indian company and everyone was talking billion dollar investments. Tesla already got a billion. Nissan was talking 1.5. GM was talking 2 billion and I said this needs a different level of investments. It's still hard to combine India and I found ourselves very good in tech. So I licensed our entire technology to General Motors. So we showcased this electric spark platform in 2010 at the auto show.

But a few, three months later GM got bought by SAIC in China and so they changed their strategy. And that's the time I spoke with Anand and initially what started as a technology licensing went to getting a majority of the company in 2010. And to me the vision was more important.

Can we build electric mobility at scale? And ran that till 2015. Built up a bunch of products, around seven products for Mandra on their platform.

Got them into also racing, so you see Mandra Racing started that business too. And then stepped down to kind of rethink because it was a long time. And that's when idea of Sun Mobility came up to start looking at kind of the questions we've been talking about, right?

Range anxiety, cost, refueling time. So I said if you could actually solve these problems there's going to be a huge opportunity. And the second part was where is the opportunity?

Is it going to be at the OEM level? I felt that the business is to more of the future going to be in energy. Can you democratize energy in a way that's quite different?

And then where do you focus on that front? And two and three wheelers were 80% almost of India. So can you give one solution for all of that?

And the second was buses and trucks, which is 2% of India but 50% of the emissions in energy. So in a way it was departure from personal cars to saying where's the biggest impact in society? Where's the largest business in energy?

And in those businesses, as an energy player, the economics made sense. If you give a customer something that was 20% cheaper to buy and 20% cheaper to operate, then there's a fundamental shift in them adopting and then there's a fundamental shift in sustainability. It came from that.

So Sanmobility started trying to look at the fact that we can make it cheaper. That was the hypothesis. I spent a year going around the world, which is a great time to take a time off, and understood why swapping worked and why it didn't work.

And trying to see the tech that needed to it. I spent three months understanding what to do in Bangalore for the buses, for BMTC, and tried to see what solutions would work and did fast charging, did slow charging, did swapping, and realized that the economics in swapping would work much better. And to rickshaws and delis.

So it was a real insight, and then the tech was very critical. So spent a year, filed my first patents, and then started a company with my partner, Uday Khemkar, who's very passionate on climate change. So what brought us together was that he was into renewables, I went to mobility.

Sanmobility is where I started racing solar cars. It's also his family side of it. So everything came together with the name and what we wanted to do is how do we run the world with this.

So Sanmobility started with that ethos of really addressing these aspects.

[Speaker 1] (1:21:22 - 1:21:25)

Can you tell us a bit more about that? What are you guys doing at Sanmobility?

[Speaker 2] (1:21:25 - 1:23:56)

So what we do is that we've taken two parts. Let's start with the micromobility. And if you sell a vehicle, sans the battery, without the battery, the cost of the vehicle can be cheaper than internal combustion engine vehicle, right?

So we heard that the cost of batteries are 30%. In some form factors, it could be 50%. In buses and trucks, it could be even 60%.

So depending on where you are, between 30% to 60%. So it would be the cost of battery. So if you know the battery, battery management system, even the charging system, then the cost of any IC vehicle, any electric vehicle, can be cheaper than an ICE vehicle.

Any form factor today. And if you look at the amortized cost of energy, battery, and the cost of energy is cheaper than fuel. And then if you think of, I can swap it in a minute, then you've addressed both the range anxiety and the refueling time.

So the concept is to create battery as a service, right? But then create it across a common solution across all platforms. So we took two wheelers use one battery, smaller rickshaws use two, larger loaders use three, and something like a Tata Ace would use four batteries.

So anything from a scooter to up to two tons, which is 80% of India, could use one battery and one infrastructure solutioning, right? And work across this form. So that was the idea around the micromobility.

Then we have solutions for trucks and buses that go from five tons to 55 tons. And we use modular batteries that can allow you to have one or two batteries to get your different range formats. Again, you can today have a truck or a bus that's cheaper than a regular electric truck or bus cheaper, and you can address range anxiety.

So if you think of in a city, you're doing your routes, right? In a city like Bangalore, 92% of your routes are under 40 kilometers, right? But you do average 200 kilometers a day.

So actually, why do you carry a three ton battery when you can carry a 600 kilogram battery? So it's 2.5 tons lighter, right? And which means your energy efficiency is 10 to 15% better.

And you can use a much smaller battery and the battery is the most expensive thing. So you cut down the cost of the battery and do a third or so, and you cut down the cost of the station because of the high utilization. And so the economics work out much better.

These are also very long life batteries.

[Speaker 1] (1:23:57 - 1:24:11)

So in scooters and trucks, scooters, they last probably around 3,000 plus cycles. So you're making the battery? Yes.

So what is battery swapping when my battery is out of charge? Do they come to your service point?

[Speaker 2] (1:24:12 - 1:24:43)

No. So today we have around 600 touch points. In Delhi, we have a little over 400.

We have a station every two kilometers. Right? By the next couple of months, we'll have more touch points than all petrol stations put together.

So it's really accessible. Within two kilometers, you can go to any network. IOCL is our partner.

So we are at Indian Oil Petrol stations here. It's easy to access, but they're at metro stations, they're independent sites, they're at franchisee partners.

[Speaker 1] (1:24:44 - 1:24:46)

So can someone like BlueSmart use this tech?

[Speaker 2] (1:24:47 - 1:26:40)

Yes, they can, but that's for cars. We haven't done it. I think in cars, for services like what they're using of shared mobility and taxis, swapping makes a lot of sense.

Anything? Definitely. We've today started the business in the two and three wheelers around shared mobility.

Because these people are driving a lot, so it makes a lot of sense because our earnings are higher. Personal mobility people drive less, so the models have to change slightly, but there's no technology issue. For example, if you look at a country like Taiwan, 92% of all electric two wheelers are with swappable solutions.

Just by one company, but it gives a solution to multiple companies. So the solution is working at scale. This year, 50% of all electric trucks in China were with swappable batteries.

That target was supposed to be achieved in 2017. So you're seeing now at scale this happening from small form factors to large form factors. So you have two models.

You have battery as a service and mobility as a service. Battery as a service, we've partnered with over 15 OEMs. So you can walk into an OEM showroom, say in this case, you want to buy a

three wheeler, you walk into a showroom, you buy the vehicle without the battery, and like you buy a smartphone without your SIM card, and then you sign up on our platform. The dealer himself sets you up.

You have an RFID card, you have a wallet, you go to any station, you can then just swap your batteries, and you're only charged for the energy you use. So you use half the battery, you get charged only for half the energy that you use. We have both pay-as-you-go models and subscription models.

These work out cheaper than even CNG as a fuel. And the cost of the three wheeler today when you walk into a showroom is cheaper than a CNG vehicle. And we do this without subsidies.

[Speaker 1] (1:26:41 - 1:26:43)

Are there subsidies for this?

[Speaker 2] (1:26:43 - 1:28:18)

If you buy it as a mobility as a service, we'll come to. I find that 10,000 crores subsidy, 1,000 is for charging. Yeah, so there's supposed to be this, right?

But again, policy lacuna and this is another big policy gap the government has, right? On this front. The second is mobility as a service, where we found a lot of fleets want to go electric.

And they don't want to deal with the infra, they just want to go. And when they cross 50 vehicles, they found it very challenging to run their fleets. So we give them the vehicle, the energy, data, and the whole service associated with it on a cost per month basis.

So today we maybe have only on mobility as a service, we have a little over 15,000 vehicles with fleet providers who in turn are using them to deliver to Amazons and Flipkarts and everyone else, right? We also have infrastructure at Amazon sites. And who's the biggest client in terms of?

They would be partners for Amazon, so they'd be like a Zip or Zingo and a bunch of other, all these. So we're on 55 fleet customers today. They may have fleets from 50 to maybe 3,000 vehicles of our vehicles on their platform, right?

And they can go to market very quickly, right? We in some months can put 1,500 vehicles in a fleet in a month, which is very difficult for them to go on, right? And now the routes change, they have no issue on range anxiety, right?

Swapping anytime they want, and it's all inclusive. So that's both battery as service and mobility as service business here, and then trucks and buses.

[Speaker 1] (1:28:21 - 1:28:25)

Is that a viable solution for you guys to collaborate with?

[Speaker 3] (1:28:25 - 1:28:36)

Swapping? Very few things we would ever disagree on. We haven't been able to agree on swapping yet.

We've been chatting for a few years.

[Speaker 2] (1:28:36 - 1:28:37)

Why?

[Speaker 3] (1:28:39 - 1:29:08)

Funnily, we were actually originally big believers of swapping. The first patent we filed at Ather was a swappable battery pack. A lot of the first early hires at Ather were all hired to build swapping stations and swappable battery packs.

What we figured eventually is a lot of practical reasons our customers weren't willing to swap a battery. What is happening is, and it's like, we can talk about cost, we can talk about tech, product, all that later.

[Speaker 1] (1:29:08 - 1:29:10)

Isn't battery inside? How do you open it up?

[Speaker 3] (1:29:11 - 1:29:32)

So, especially building a more premium vehicle, what I kind of realized is, and I remember having this conversation back at home with my family. Because the idea initially was very powerful. It totally is.

It's very hard to not fall in love with it. In fact, I think I raised my series A funding by saying it's swappable battery pack.

[Speaker 2] (1:29:33 - 1:29:34)

We'll get you there, you're going to all turn around.

[Speaker 3] (1:29:35 - 1:30:11)

Let me hear you out first. I'm acknowledging that I have believed in it for a long time. The early designs were actually all with swappable battery pack.

We thought of robotic stations and everything. What happened is, I had this chat at home, and

everybody liked it until I asked them, until they asked me, how will the swapping happen? And I said, you will take the battery out and you will put it on station and put another one out and put it in.

Oh yeah, makes sense. And how heavy is the battery? And then we ran into issues.

So I said obviously we'll break the battery into two parts.

[Speaker 1] (1:30:12 - 1:30:12)

How heavy is the battery?

[Speaker 3] (1:30:13 - 1:33:18)

20 kgs. So we'll break the battery into two parts. So you'll only have to lift like 10 kgs at a time, max.

Maybe 8 also. And I realized nobody was very excited about it. And very quickly it was like, Oh yeah, fine.

Papa will swap it. I can't. Mother can't.

But dad has a back issue. So maybe sister will swap it. But she's not in Ahmedabad right now.

So Tarun, if you go, you'll swap it. And I very quickly realized the only way to actually solve this problem because most Indian consumers were not willing to do a manual swap. See the example that Chetan talked about, very powerful.

Taiwan. The Taiwanese customers were willing to do a manual swap. They were willing to lift up that 8-9 kg battery, one half, 8 kgs, pull it up, slide it in a rack, pull on the 8 kg cylinder out, slide it in without breaking anything.

Do this twice every couple of days. But I realized that the Indian customer, personal consumer, like absolutely, fleet consumers will do it in a heartbeat. It's a no-brainer.

Personal users were not super excited about doing it. And the only way to run the entire solution for them was to have an attendant at the station who would do the manual swap for them. But the minute you introduce an attendant cost as a company in this entire infrastructure, everything just stops making sense.

That's what happened to us. Then we ran into a second issue, again not cost-related. So we ran this very small experiment.

Basically I would do a lot of these mock protos. Have the consumer just walk through the experience and see at what point do their eyes break contact? When do they switch off?

And as walking through somebody this experience, I realized the person was I just bought this scooter. Brand new scooter. On Sunday.

Monday I go to a swap station. And I'm going to change my battery. When do I get my battery back?

I'm like, never. Your battery is a commodity. Just think of it as petrol, right?

We'll give you a battery. It's fully charged. Full range?

Not full range because obviously life will deplete. You will not always have a 100% battery. After 3 years it will be at 80%.

But you will get 80-85% generally. That's not fair, no? I just bought a brand new scooter.

You told me 100 km of range on Sunday. On Monday you give me something at 85 km of range. A 3 year old battery.

And I was just playing. It has a junk, a dent. I want a new battery.

I bought a new scooter. I realized I can't solve this problem. It is possibly solvable, right?

But I can't solve this. I have to fight so many other battles with the Indian consumer today. Buy an electric scooter, buy a premium scooter, buy a connected scooter.

If I add that you don't own the battery. Your range will be plus minus 20%. And you will have to do a manual swap.

I realized like I'm just guaranteeing our business will fail. The way we are set up. So we pulled out of it.

Which is why we haven't agreed yet.

[Speaker 2] (1:33:19 - 1:37:10)

So counterpoint to that. All valid stuff. So we started with shared mobility.

you know. Where there is not an issue at all, right? And when you start with personal mobility you know and you are selling your first 100 scooters.

You can't put infra all around the city. Because the guy is going to say where do I have it. But once infra is in place 500 stations in Delhi, right?

We have 100 in Bangalore. 85 LPG stations in Bangalore. We have 100 catered to 200,000.

We have 100 stations here. Suddenly you are saying hey I see the one outside my office. I see

one outside my work.

I see one outside my kids school. Now I'm thinking differently. So for us it's a journey.

You start with shared mobility. Because there you are putting 500 vehicles somewhere. You put the infrastructure up.

Once you have enough infrastructure then there is an option to look at personal mobility. There are several other things that we have learned also along the way, right? One is that we are doing a million swaps a month.

So there is enough consumer experience. I probably agree with him that for personal mobility you need slightly lighter batteries. You have different segments of the population and shared mobility is more willing to do it.

But that's okay. It's not a technology issue. It's just what you would adapt to for that application.

So that's not an issue at all. The second thing is that if you walked into a showroom and the customer you had a scooter that was 1.5 lakhs or so. It would probably be 75 to 80k if you remove the battery, a 4 kilowatt battery if there was that and a charger and everything else.

Now you could tell the person do you want to buy something at 75 80k or 1.5? But the performance is identical. The speed is identical.

The acceleration is identical. The connectivity is identical. You can't even tell the difference between these two scooters.

What would you buy? Now let the consumer make that decision. What we found with a lot of OEMs is they have a petrol, they have a diesel, they have a CNG, they have an LPG, they have an electric with swappable and they have electric and fixed.

And in some markets 70% of the sales are electric with swappable. Which means consumers are walking into a showroom, have options and decide. Now 80% of India doesn't have infrared home.

Today when you start in the premium market you're capturing the 10%, then the 15, then the 20. But once you go, if I ask a staff in here, where do you go, where do you park? My mage says I bought a new scooter but I park 200 meters away from where I have to go and I don't have infrared so I'm not buying electric.

So as you start to go to scale, infrared becomes an issue. And that's when the swapping really adds it because it's one minute to do it. So it solves that problem for you.

The third reason in my 15-20 years is everyone was concerned what happens to battery life. I bought this scooter, I bought a vehicle, it's giving me a three year warranty. Maybe an extended

warranty of five.

I have to pay a lot more against it. But if it goes bad, I have a huge expense towards it. And unfortunately because of the taxation and everything else in India, a battery replacement cost could be 30% more than an OEM cost structure.

So that really adds a huge cost. So this battery could be 80,000 plus for a replacement battery. When someone hears that they say, are you telling me I have to spend that much money that period of time?

So that's another deterrent why they do that.

[Speaker 1] (1:37:11 - 1:37:23)

On the other side, Can I ask you a question? If you have the inventory of batteries that you're swapping now and let's say the life cycle of a battery is 8 years, 10 years, whatever the number might be. It's much less for us.

[Speaker 3] (1:37:23 - 1:37:28)

The reason is, we swap a lot more.

[Speaker 2] (1:37:29 - 1:37:39)

So our batteries are swapping twice a day. Our batteries are 3000 cycles, they last us 5 years. But we're using it a lot more every day.

So it's a long life battery.

[Speaker 1] (1:37:39 - 1:37:41)

So what do you do with the inventory as a business?

[Speaker 2] (1:37:42 - 1:37:47)

We have around 1.2. So for every battery in the field, you have around 20% in our stations.

[Speaker 1] (1:37:47 - 1:37:51)

And beyond 3 years, will everything be running at x% of the efficiency today?

[Speaker 2] (1:37:51 - 1:38:22)

So our customer promise is that it's 80% you'd always get that much. But there's another interesting insight. We've 6 months ago introduced a battery that's 45% more capacity.

A customer that's loving it because they've over the last 2 years had range and now they have 40% more range. 45% more range. They're paying on the same amount per kilowatt hour.

So for energy they're paying the same, but they have performance.

[Speaker 1] (1:38:22 - 1:38:25)

How do you monetize this? Who is paying for it? So you're doing B2B.

[Speaker 2] (1:38:26 - 1:41:53)

Yeah, so as B2B and B2C. In B2C, you're going in, you're using a certain amount of energy, you pay per kilowatt hour of energy you've used. If you're B2B, I've included that.

So you're not paying per swap? You're paying per swap. Or are you paying per kilometer driven?

You're paying per energy used. So think about it. Kilometer driven, I'm giving you one liter of petrol.

You may drive at 15 kilometers per liter. Someone may drive at 10, someone may drive at 20. So I can't give you on a kilometer per liter.

I can give you as a unit of energy. So like rupees per liter, I'm giving you rupees per kilowatt hour consumed. We're digital and it's to the third digit because it's all electronic.

So exactly how much they've used comes on their screen, comes on their app, comes on their SMS when they get their invoice, and that's how much they've used. They only charge for that much. So that's this part of it.

But what's very interesting is we've had customers who've done 1.5 lakh kilometers on their three-wheelers. And their friend bought one with a fixed battery and after three years they had to spend a lakh of rupees in replacement. And the battery capacity started at 100 and in three years dropped around 65%.

Because in fleet usage you use a lot more than probably personal. In this case, this customer after three years has got a battery that's 40% better. Now in two years, a year from today, it's going to get another 10%.

So in the 10-year life cycle of their vehicle, their vehicle has gotten better every year. Whereas you're buying something, a depreciating asset that goes bad. So when you start to think of all of this, it's not a silver bullet.

Every solution is going to co-exist. Slow charging, fast charging, swapping are all going to co-exist. It depends on the customer and their choice of what they'd like to do.

Do they have infrastructure at home, they don't mind paying more, they don't like to go somewhere else, they like to charge at home. That would make sense. So each one depending on the customer's requirement.

But finally the cost structures are far more beneficial. And also the benefit of never having to worry about the battery. When someone picks up a scooter or any vehicle, my big insight has been that they buy the vehicle, when they buy the vehicle, they always think of 1% of their usage, not 99%.

99% of the time, they're doing less than 40 kilometers a day. That 1% of the time, they've got an extra trip for school with their spouse, an extra call, and they may have done 100. And when they do 100, they're looking at least having 120 kilometers.

And then they de-rate whatever the OEM says, unless it's Ather, where you're actually the other way around, where Ather gives you a very good, I mean they're very transparent, which is very good, they're building that up. But that's an exception to the norm in the industry here today. So now suddenly if you want that much range, you probably need a 4 kilowatt hour pack for a scooter.

And even at $120 a battery pack, that's $480, $500, with charger and everything, it may be, you know, it's more than 50% of the bill of materials. So wherever it goes, it's going to be a very large chunk on this front. And then the other challenge in scooters and three-wheelers in India, is that you can't cool them.

Very difficult.

[Speaker 1] (1:41:54 - 1:41:55)

You can't have air conditioning.

[Speaker 2] (1:41:55 - 1:42:01)

Is it the batteries that you can't air condition? Well, the battery life, you know, at 45 degrees, the battery life could be maybe a third or half of...

[Speaker 1] (1:42:01 - 1:42:04)

And these batteries, are they imported or are they made in India?

[Speaker 2] (1:42:05 - 1:42:42)

The cells are imported. Most companies would import the cell and make the packs. Ather does that, we do that, right?

I think in the next 24 months cells will start being produced in India, right? And maybe in 48

months we'll have a reasonable chunk. But still, a lot on the cell side, like what you talked about, you know, the panels and the cells, right?

But that value is pretty high for a two-wheeler because the packs are smaller. So that added value could be 40% or so to 30 to 40%.

[Speaker 3] (1:42:42 - 1:42:46)

It's lesser now. Much lesser. It's approaching what cars were at Model S.

[Speaker 1] (1:42:46 - 1:46:15)

Actually, I should have framed that question to you and not to you. He's more B2B, right? Swapping in our case won't work.

I'll tell you my thought. Cars are premium products. People who buy cars, they want, they're not the ones going to pull out a 40 kilowatt hour battery or 60 kilowatt hour battery of the car, right?

They want ease, they want comfort. That's not a two-wheeler. Nobody, whether a fleet operator like BlueSmart or an individual person, wants the discomfort of swapping a battery.

Because you bought a car for a reason, right? And your cars don't have utility for 24 hours. I'm just setting the context coming there.

As more and more global OEMs I speak to in the US or in Europe or in India, they're all of this thought process that I mentioned that nobody wants to probably replace a new battery with an old one. They were not so attracted towards building a car with swapping, where you take the battery out and change. The good part about, the good news for us is that the range of the car is going up, right?

The range of the battery is going up. You don't need today, I mean for intra-city purposes, you don't need batteries which are more than like 80 kilowatt hour or 60 kilowatt hour storage. That will give you nearly 500, 700 kilometers.

I mean we have companies which are today or large brands which are promising 700 kilometers or 600 kilometers on single charge. But you take the consumption, it doesn't happen with personal mobility. Even if you look at fleet operators like BlueSmart, for intra-city purposes, within the city, even if NCR you see, I mean you can't exhaust more than 300 kilometers.

So swapping doesn't make sense for us. I mean that's what our thought is, globally. I'm just telling our thought process of other fleet operators that we speak to in the US with also using Tesla Model Y cars for running a fleet like BlueSmart.

They are equivalent to BlueSmart in the US. Best part about is now the cost. So if I were to now go back to unit economics and say look you know, Apple to Apple, if I bought a Civ Desire for 7-8

lakh rupees, I mean now the new Civ Desire apparently cost vendor for 6.5 lakhs to 8 lakh rupees. Today you can't get a Civ Desire below 8 lakh rupees, a good loaded one with BS6 norms. A Tata car, I will have to shell out 4 lakh rupees more, 3.5 lakh rupees more on the Capex. So you are saying that look my Capex is high here, it comes with a battery, my Capex is low here, it doesn't have a battery here right now.

But look at the utilization, if I were to use 1 lakh kilometers and pay on an average between CNG and petrol and diesel, a mix of about 6 rupees per kilometer, I'm shelling out 6 lakh rupees on fuel for running the car for lakh kilometers which is the consumption is about 14 months or about 15 months. The same consumption on electric, I will have to pay a footer bill of 80,000 to 1 lakh rupees. So the additional amount I paid for the car, I got back in 14 months just in the fuel cost.

So the additional amount I paid for buying that electric vehicle with that additional battery, I recovered that in just 14 months because my operating cost was low. So I'm happy to pay for that battery up front and utilize that over 14 months and then I'm on par in 14 months time. And then I can continue to run my car, charge my car at ease as per utilization.

But swapping for four wheelers, for cars I think Can I ask a question? Without battery swapping, if you want to transport something from Bangalore to Goa which is 1000 kilometers, how do you do it?

[Speaker 3] (1:46:16 - 1:46:36)

I mean you need to have charges in between and stop for example. And how long will it take to charge a truck? A truck will likely have 500 kilometers of range.

You will stop once for about 3 hours. You will do not a full fast charge, you will do basically a hybrid fast charge. Typically when these buses, trucks charge they're using 3 hours to charge the battery.

[Speaker 2] (1:46:38 - 1:48:19)

And what you're doing is you're by swapping over 16 hours, you're using an energy over 16 hours where each truck would use it over 3 hours. So in a way you think about it, the grid power that you need, peak power is almost 3x more. And we found this when we did a pilot in Ahmedabad with 18 buses with fixed charging and swapping.

The buses with fixed charging would charge over 4 hours at night. They would swap the charges out for us for the other buses. And the same buses were taken care of at one station.

And we needed around a megawatt of power and they needed 2.5 megawatts of power. As you go to this scale, a megawatt of power a megawatt of transformer cost will cost you a crore rupees. So you have to think holistically.

Then you need parking for all of these buses. So I mean when you have a huge place today to scale and you have 5000 vehicles or 50,000 vehicles, you have to park 50,000 vehicles. What is the real estate cost of 50,000 vehicles versus you come in, you swap it.

Each of our stations for the two wheelers today do 200 swaps a day. So you have 600 stations so on a daily basis I can cater to 120,000 scooters. That's the kind of throughput we have.

So the scale you're able to do to add a million vehicles in a country and 5 million vehicles, your infrastructure, the energy cost, the real estate cost, the solutioning around, the whole thing has to come together. Also it kills a bit of range anxiety doesn't it? We find that tremendously, we find that at least that when someone knows that two kilometers away there's something, they use more of the battery you know.

They also gun it.

[Speaker 1] (1:48:20 - 1:48:25)

And he said eventually the differentiation is going to be brand like it is for ICs.

[Speaker 3] (1:48:25 - 1:48:27)

Every business, ever.

[Speaker 1] (1:48:27 - 1:48:29)

It's not range, it's not this, it's not that.

[Speaker 3] (1:48:29 - 1:48:33)

It's all commoditized. I just have a quick question.

[Speaker 4] (1:48:33 - 1:48:41)

Are you saying tier one, tier two, tier three, everywhere in India the same rules apply? Range anxiety, charging.

[Speaker 3] (1:48:41 - 1:50:38)

So I'll give you usage data. We now sell across 100 cities so it's reasonable tier one, tier two, tier three distribution. Our users are across all three towns, equal numbers.

I think tier one daily usage is 31 kilometers, tier two is like 32, tier three is like 29. There is no meaningful difference of usage, daily usage. There is no meaningful difference of buying patterns.

Tier one, tier two, tier three. That's surprising, you're saying people in tier three are buying the

same percentage as tier one? It is very confusing.

Why is that? Everybody is buying in the same proportion of the premium and the entry level. Why is that?

I think because everybody is generally misjudged our economy. India has always been supply constrained, not demand constrained. Like in two wheelers also, my biggest belief has been since we started Aita, that everybody is misjudging the market by trying to build a, like trying to build a cheaper and cheaper and cheaper two wheeler for a long time.

I bought a, or rather my friend bought a Honda Activa when I was in 10th standard, when it just came about. First Activa's came about in the country. It was I think about 45,000 rupees back then.

That's 2005. To 2015, when we had started Aita, or actually I forgot, 2020. 15 years later, the damn product was still only 75,000.

It didn't grow with inflation at all. Now it is 90,000. Now it is 90,000.

Now it has been affected a lot by COVID. But, the consumer has moved like a lot of steps ahead in life. That same customer, when he goes to the market, he's like, if I'm buying a bike, I'll spend 1.6 lakhs. If I'm buying a scooter, he's only spending 90k. Why? It's not like the money is over after buying a scooter.

No one is making a scooter for 90k. I had always believed this. We kind of hedged our bet launching multiple variants, but every damn time we've been surprised.

[Speaker 1] (1:50:39 - 1:50:50)

Before I finish with Chetan, for our crowd, for an entrepreneur young, starting off, what is the opportunity in battery, like battery swapping? What can somebody build?

[Speaker 2] (1:50:52 - 1:51:58)

So, see, the market is very large. There's also an existing market of e-rickshaws. There's around 2 million e-rickshaws, which in battery swapping and energy business is around a billion dollars a year.

They're all in smaller cities. And so you've had a lot of smaller startups creating solutions for retrofit kits for these e-rickshaws because they use lead-acid batteries and after nine months, these batteries die and they have to replace it, which costs around 35,000 to 40,000 rupees. So you could, at that point, get into a subscription model and they're able to then also use it in, you know, they do it during the day, they give it to their nephew or someone in the evening for three hours, so they can use it and let acids take five, six hours to charge, unlike lithium ions, right,

because they don't have that.

So clearly there's an opportunity. And because it's going to be different, India's such a big market, it's going to be difficult to be in every market and every place.

[Speaker 1] (1:51:58 - 1:52:07)

There are going to be opportunities for micro entrepreneurs to look at such options. Set up charging stations or battery swapping stations.

[Speaker 2] (1:52:07 - 1:52:20)

So they can be in the business of battery swapping, right? In the battery side of it, I think there's also, you know, we're doing franchises, and so there could be a franchisee who takes a station and they get a certain return.

[Speaker 1] (1:52:20 - 1:52:28)

How would that work financially? How much money would I need to put up to be a franchisee? What is it called, your battery swapping company?

The Sun Mobility, right.

[Speaker 2] (1:52:28 - 1:52:43)

So if you're looking at a smaller station, that'll be around, well, that'll be launched in a couple of months, it'll be a lakh and a half as a franchisee fee. And Capex? That would be a franchise.

So that's what you'd give, it's a deposit, and then you get a certain revenue share.

[Speaker 1] (1:52:44 - 1:52:48)

And then, you know, Do you have to spend to set it up, the store or the space?

[Speaker 2] (1:52:48 - 1:53:44)

The smaller one, they can set it up, we just send you on wheels and it gets all online, right? The larger stations need a little bit more, the investment is around five lakhs for that area. This is not including real estate and all of that?

No, so someone already has a shop next to that, they put this, and then they have extra business, right? And then they also have secondary businesses, maybe a tire shop, and he does this. Or he may be having other goods to sell.

So not only has the revenue from this, but he also creates that. So I mean, all our dealers have that, right? So if you think about OEMs, dealers have this.

If you think about it, five years from today they're not going to have revenue on service as much because the service of electric vehicles is much less, you know, oil changes and all this stuff. How will they make revenue on an ongoing basis? So by giving them a franchising model, they are able to make revenue.

Not only that, typically a station may see 50 customers a day. So those 50 customers that are coming, you can cross sell them and say, hey, oh, your vehicle needs service.

[Speaker 1] (1:53:45 - 1:53:48)

And how much do you mark up the electricity, your cost to the recharging thing?

[Speaker 2] (1:53:49 - 1:54:22)

I don't think it's, think of it as a markup. What you're doing is you're taking a battery and you're amortizing it, right? And then you're adding the charging infrastructure and amortizing it and then the energy and amortizing it.

But let's take a scooter. A petrol scooter would probably cost you around 2 rupees, 2 rupees 20 paisa per kilometer, right? A swap scooter may cost you between 1 rupee to 1 rupee 20 paisa per kilometer, right?

And the cost of scooter would be cheaper than a petrol scooter. I'm just giving you an idea of what that would mean on this front end.

[Speaker 1] (1:54:23 - 1:54:40)

So if I were to put up 1 lakh rupee and set up Sun Mobility battery swapping next to my store, I'd probably get between an 18 to 22 percent return minimum. Even today? Even today, yes.

At the frequency of people using battery swapping?

[Speaker 2] (1:54:40 - 1:54:44)

And they could be higher in the years to come as you start to look at it.

[Speaker 1] (1:54:45 - 1:54:51)

So lastly, Tarun. So Tarun is a bro and we hang out.

[Speaker 3] (1:54:51 - 1:54:55)

We recently spent I've been dragged into the hanging out.

[Speaker 1] (1:54:57 - 1:55:12)

So I know his personal life a bit more than others. Also, I have researched Tarun's company Ather a fair amount. Maybe Tarun can start telling us a bit about himself from the beginning for everyone else listening.

[Speaker 3] (1:55:13 - 1:56:33)

I'm a Marwadi. Can't speak though. But born and brought up in Ahmedabad.

But can't speak Gujju either. One of those guys who had to go to science because marks are there. So he'll take science.

And that science, somebody suggested you should write JEE. So went for coaching. Cleared JEE luckily.

Got a seat in IIT Madras. So went there. Did a 5 year course there.

Graduated in 2012. And we started Ather a year later in 2013. Our love, I was trying to say even in the start at some point, was actually our love actually was the energy industry.

From like a very long time. Me and my co-founder Swapnil have been calling ourselves Ather Energy since many many years. So we really always wanted to start an energy company.

Which is why we called ourselves Ather Energy and not Ather Motors. Thanks. And to this date we keep hoping that this becomes an energy company before we die.

Explain how energy company? It's not. It's not technically an energy company.

How I don't know. But life is in some ways very short and in some ways long enough. So I kind of think 10-20 years to build a two wheeler business and then another 20-30 years to build the other stuff.

[Speaker 1] (1:56:34 - 1:56:44)

Do you think at some point with scale it would make sense for you guys if you own the charging infrastructure to also generate your own electricity maybe using solar?

[Speaker 3] (1:56:45 - 1:57:39)

Actually that's the plan. The energy cycle is basically generation, distribution, storage and then application. In college we spent many years trying to work on generation.

We tried to build engines. Very very high efficiency engines. We succeeded technically not commercially though.

And later on the idea that we had was to build battery packs which is storage. And we wanted to

build a swapping network which is distribution. So we justified to ourselves by saying that we're not working on generation but we're working on distribution and storage.

This is Ether Energy, the distribution and storage company. And later we built we were building a scooter on the side to put that battery pack on and we fell in love with the scooter. So we became a vehicle company.

But I kind of still think like we do run a fairly large fast charging network. In two wheelers it's the largest fast charging network in the country. How is that?

[Speaker 1] (1:57:39 - 1:57:45)

Because your competition has incumbent distribution through service centres, right?

[Speaker 3] (1:57:46 - 1:57:48)

But they don't have charging there, no?

[Speaker 1] (1:57:48 - 1:57:57)

But I would assume it's easy for a TVS or a Bajaj or a Hero to add a charging infra to existing distribution.

[Speaker 3] (1:57:58 - 1:58:56)

So the problem is not technology, the problem is what business are you in? And it's just the lens that you apply. If you believe that you are a vehicle company as defined by the definition of we put together a vehicle and we sell the vehicle then getting into the business of charging unless forced to seems kind of odd.

It's like if you make a car, let's sell petrol as well. How difficult is that? Our customers, captive customers.

But you don't, right? Because you kind of understand that that's not our business. Companies understand this much.

So they don't get into charging unless they will really be forced to. We just think differently. We actually see this like as Puneet was saying plus energetically.

Listen like, we'll make a car, we'll make our own battery, we'll do the charging ourselves. We'll install enough charging stations and if you fast charge there then we'll need storage there that will supply the power to fast charge these batteries eventually. You set up like 20 stations of charging that becomes like a micro grid.

Then you've got decentralized power generation, distribution, storage. There are a lot of dreams.

[Speaker 1] (1:58:57 - 1:59:29)

You know classical economic theory like Adam Smith or the Austrian theory of economics. Take the example of a needle. When somebody tries to make a needle from start to scratch it's likely less efficient than somebody who is mining the metal required for the needle.

Somebody else having another expertise. So why is, why are both of you in particular when it comes to your particular companies trying to do everything from start to finish?

[Speaker 3] (1:59:29 - 2:00:16)

No, not everything. There is a business. So there is a business.

It would be weird if I said we should build our own tires. You're not in the business of building tires. So if I said my business is building vehicles then thinking charging, thinking batteries is kind of not sensible unless you're forced to.

But if I think I'm in the business of selling you energy as it really is, then the lens changes. And then a few things start making sense. But if you're not in the business of selling energy today, why are you thinking on these lens?

So that's a dream. I think some of the things, most of the stuff that we do today, including setting up charging stations is really in the service of the vehicle that we have to sell.

[Speaker 1] (2:00:17 - 2:00:20)

Are these brand agnostic? Can anybody go to your charging station?

[Speaker 3] (2:00:20 - 2:00:37)

Our charging stations, a lot of people can now go because our charging standard has become the national standard. So if I have an Ola scooter, I can go to a charging station. Because Ola hasn't adopted our standard yet.

But Hero Veda has, Matter has. A bunch of e-two-wheeler companies and startups and charging point operators have.

[Speaker 1] (2:00:37 - 2:00:39)

Also because Hero is your investor.

[Speaker 3] (2:00:39 - 2:01:01)

So that gives us an advantage there. But it also brings the largest two-wheeler guy on the standard platform. So the company starts thinking this way.

How can we get more people to adopt a standard? You won't think like that if you're a two-wheeler company. If I think like a two-wheeler company.

But if I think like I'm ultimately, like 50 years out, I'm the business of getting you low cost energy. Then I'm thinking a little differently long term. My decisions are a little different.

[Speaker 1] (2:01:02 - 2:01:11)

So market shares today is say 30 Ola, 20 TVS, 15 Ather, 12 Ather. 7-8 Bajaj.

[Speaker 3] (2:01:12 - 2:01:13)

Bajaj is also at 12-13 now.

[Speaker 1] (2:01:14 - 2:01:15)

So Bajaj has picked up.

[Speaker 3] (2:01:16 - 2:01:19)

They've got enough distribution today, right? So they can turn on the distribution. They can open shops everywhere.

[Speaker 1] (2:01:19 - 2:01:28)

So, with distribution, these guys are able to achieve scale faster than an Ola or Ather might be?

[Speaker 3] (2:01:29 - 2:02:51)

Short term? Absolutely. But why not long term?

Long term distribution is not really a moat in India. It's not really a distribution. Because that was the 80s India.

When MSME was a difficult thing. So if you could get enough entrepreneurs to sell your vehicle, you have actually a moat. But today, MSME interest in setting up E2 wheeler distribution is crazy.

We have, I think, this is 6 months ago, we had I think 20 or 30 thousand dealer applications. For a business case that requires you to put 2 crores upfront. Now there are a lot of people who could do this.

There are like 10,000 companies who run automotive dealerships in India. All scared to hell. Otherwise we have no future.

So what are the best E2 wheeler companies or electric car companies? You are already coming with an investment mindset. I want to invest and learn this for the next 5 years.

So you can scale up. So for example, we've gone from like 0 to 100 touchpoints in like 2 and a half years. We'll go from 200 touchpoints to 1000 touchpoints in the next 2 and a half years.

At which point you've crossed oil and fuel scale already. What took them possibly 30 years. So distribution is a very weak moat now in India.

So it's a low hanging advantage for incumbents. Because they can turn on the tap faster. Maybe a year ahead of us.

But it's that much. After that it's a lot of other things.

[Speaker 1] (2:02:52 - 2:03:20)

I think I keep coming back. Here I have like a personal interest. Especially in your space, more so than yours.

Because it's more competitive in the public domain. What is happening to Ather, what is happening to Ola. Because there are 2 very similar companies trying to build a similar product.

What do you think Ather is doing well where Ola isn't? And what do you think Ola is doing well where Ather isn't?

[Speaker 3] (2:03:20 - 2:04:03)

What Ola is doing well and what we are not is getting the message out. Marketing? The marketing is damn good compared to us.

They've been able to crack. They've been able to ensure that a lot of people have heard about them and the fact that they have a 2 wheeler. Every brand track, every marketing research that we do.

The biggest problem that we discover is that people don't even know what is Ather. Like up north, the biggest feedback that I get is that most people think that if they have heard of Ather it must be some European Norwegian company. I don't know how long it will stay in India.

What does Ather mean? It's a Greek word. It's the cleanest form of air.

So Ather Energy was meant to be the cleanest form of energy.

[Speaker 1] (2:04:03 - 2:04:13)

What is the difference between a cell and a battery? And when we said batteries all of these times and in the Indian context, what does it actually mean? Do you want to take that?

[Speaker 2] (2:04:13 - 2:04:56)

Sure. We see battery as a battery pack which is looking at cells. Then you integrate thermal management, the mechanical.

Put all the sensors, the electronics and then you're closing it up and making it a pack. There's a lot of hardware and software and other stuff that goes. It's customized for the application.

It's designed for. And the temperature as well. Thermal management is very important.

The new regulation coming in. There are a lot of new safety regulations that have come in that are quite stringent. That require a lot of innovative designs to manage thermal on the side for safety as well as to enhance your life on this front.

[Speaker 1] (2:04:56 - 2:04:58)

Is anybody making cells in India?

[Speaker 3] (2:04:58 - 2:05:45)

Are we able to compete with China? Making cells is a commodity. It's not a technology business.

It's setting up any commodity manufacturing today. The challenge of setting up cell manufacturing in India for a while now has been demanding here. And the minimum investment required is pretty high.

If you're not bringing 4-5000 crores to the table practically, it'll be very difficult for your plant to have the size and your purchasing to have the size to really make this make any sense. Forget competing with the LGs and Samsungs of the world. But the other problem has been demanding.

Even if you set up a 5000 crore plant, the production out of it has to be sold to somebody. That's only happened now. Is Ola attempting?

Ola is attempting building the cells.

[Speaker 1] (2:05:45 - 2:05:46)

Why is that?

[Speaker 3] (2:05:47 - 2:05:51)

The logic there is that vertical integration will help reduce cost further.

[Speaker 1] (2:05:52 - 2:05:54)

But will that take the cost below Chinese?

[Speaker 3] (2:05:55 - 2:06:35)

I'll give you one data point. Most cell manufacturing companies, the really enormous ones like LG Chem, Panasonic, all these guys, even in the best times that they've made cells and sold cells at like crazy ability to sell price at any point. They can sell at any price.

Like last year basically. Their pads have generally not crossed into double digits. They've been generally like 4-5%, 6% pad businesses.

And they're very high technology businesses. So I've generally considered cell manufacturing as this really complexly bad combo. Very high tech with commodity margins.

So it only makes strategic sense. It generally doesn't make cost sense.

[Speaker 1] (2:06:35 - 2:06:57)

I'm trying to understand as a layman, the chemistry of a cell. But maybe we start from the very simple fact of cell, battery, what is it? What is involved?

I don't know who wants to take it. Very simple layman, like think I'm an idiot, but look at me and explain it to me, not to you guys.

[Speaker 3] (2:06:59 - 2:07:03)

Then you should take a stab at it. You'll explain better than most of us on the table.

[Speaker 2] (2:07:06 - 2:07:44)

So a cell has an anode, and a cathode, and a separator that separates the two so they don't short. And then you put an electrolyte in it. And then you put that into a container and then put the electrolyte in it.

And that's what forms, think of your typical AA cell or anything that you have. And cylindrical cells are very common. They're still a larger size, something closer to probably something between a C size battery.

Approximately that. It's got a 26 size diameter and a 750 in length. So this would be probably a standard way of how you look at a cell.

[Speaker 3] (2:07:45 - 2:07:51)

It's literally filled with, it's called a jelly can, jelly roll. It's literally filled with jelly. That's the electrolyte layer.

[Speaker 1] (2:07:52 - 2:07:56)

So lithium and cobalt does not come in at the cell level. It comes in when you package it.

[Speaker 3] (2:07:56 - 2:07:57)

No, no, no. It's in the cell, right?

[Speaker 2] (2:07:57 - 2:09:14)

So your cathode has your, it has an NMC, which is nickel manganese cobalt. It could be NCA, right, on this front, which is what say some companies do. And then you have LFP, which is lithium manganese phosphate.

So there are different chemistries So when I hear lithium ion battery, what is it? So it could be all of them. So lithium ion would be anything that is a lithium battery.

So what is the role of lithium in a cell? It is one of the materials, I mean if you think of a cell to create a potential difference, right, you try and get two materials, and if you remember the periodic table, at an end where you can get a big difference. Lithium happens to be an area with a potential difference between that.

And the energy it can keep is very high compared to, say, lead. So lead, aluminum, aluminum, lead acid batteries, right, your standard batteries, they all have different nickel zinc. Each one has this.

You're taking two materials that have a differential in the periodic table and you're using that to look at it, right? So that's the more, you store more energy in a battery if it can, like lithium, because the difference in the periodic table is much higher.

[Speaker 1] (2:09:14 - 2:09:17)

I know nothing about science, but when you say differential, what do you mean?

[Speaker 2] (2:09:18 - 2:09:38)

Well, think of it that if you had water and you try to put water from one feet or three feet, right? So think of it from three feet, you'd get a lot more, versus from one foot, right? So that would be the potential difference you could have.

In? In, that would be the equivalent of saying I have two materials, right?

[Speaker 1] (2:09:38 - 2:09:45)

And if the potential is higher, right? When you're saying potential, I'm not, my mind is not able to picture potential of what?

[Speaker 2] (2:09:46 - 2:10:03)

So the potential would be, in this case, let's simplify, would be voltage, right? So if I have a bigger

difference between two points, the voltage would be bigger, and that's equivalent for me, you know, pouring water from a foot or three feet, right?

[Speaker 3] (2:10:04 - 2:10:05)

So if I'm pouring it from three feet, there's a lot.

[Speaker 1] (2:10:05 - 2:10:09)

Are you talking about the material's ability to pass voltage through it?

[Speaker 3] (2:10:09 - 2:10:11)

No, that's resistance.

[Speaker 1] (2:10:12 - 2:10:17)

So when you're saying potential, potential of storing electrons?

[Speaker 3] (2:10:17 - 2:10:46)

No, it is the, potential is basically the difference between the cathode and anode, right? So if you've got a bigger difference, electrical difference between them, then you can generate, with all things being equal, a higher amount of current, okay? So current, think of that as the speed of the water, right?

That's it. So the higher up you go in the building, the faster the water will drop, right, when it comes to the ground. If you drop it from, as you said, like, stable height…

[Speaker 1] (2:10:46 - 2:10:48)

So you're saying if the battery is bigger…

[Speaker 3] (2:10:48 - 2:11:01)

No, it's not the length. It is, it is the electrical difference. It is basically the periodic table, what he's talking about.

Usme, the difference comes in. So lead will generate lesser potential, and lithium will, I think, generate higher potential.

[Speaker 2] (2:11:01 - 2:11:08)

So if you take the standard cell that you get for your remote control, it has 1.5 volts. That's zinc, right?

[Speaker 3] (2:11:08 - 2:11:09)

Right, right.

[Speaker 2] (2:11:09 - 2:11:45)

So, zinc carbon, or whatever. Then you look at lead acid, right, it will be around 2 volts. Then you look at lithium ion phosphate, it'll be around 3.2 volts. And you look at lithium ion or NMC chemistry, it'll be around 3.6 volts. So now if you think that I can have 3 times or 2.5 times the potential voltage, which means the amount of energy I can get is much more because energy is current times the voltage. So if I have 2 times more voltage, 2.5 times more voltage, then I can give you a lot more energy. So is this where lithium and cobalt is used?

[Speaker 3] (2:11:46 - 2:12:04)

Cobalt is, see, basically, you dope these things. The voltage difference is not because of cobalt. You can have cobalt or low cobalt chemistries and the voltage will be very similar.

Cobalt determines the energy content and the I think the resistance is different because of the cobalt that you use.

[Speaker 4] (2:12:04 - 2:12:12)

What are the ions actually going across the anode? Is it actually the lithium? Is it lithium ions which are moving?

[Speaker 5] (2:12:12 - 2:12:12)

Yeah.

[Speaker 4] (2:12:13 - 2:12:17)

So surprisingly, somehow I feel like I should know this, but I don't.

[Speaker 5] (2:12:17 - 2:12:19)

You're the one who's the PhD.

[Speaker 4] (2:12:20 - 2:12:26)

And yet, is lithium like a consumable? So what happens at the end of a battery's life?

[Speaker 3] (2:12:26 - 2:12:49)

When you charge it back, it goes back, right? So at this point. So over time, the lithium is not consumed.

A cell loses life because its resistance keeps increasing. So what happens is the medium through which the medium through which electrons are travelling, no? That just keeps getting, for lack of a better word, just worse and worse.

Just more more the resistance is increasing.

[Speaker 2] (2:12:49 - 2:12:51)

So over time, there's deterioration.

[Speaker 3] (2:12:51 - 2:12:54)

Electrons don't flow well. There's deterioration with what's called lithium plating.

[Speaker 2] (2:12:55 - 2:13:19)

So if it goes, the voltage goes very low, that the lithium is actually deposited. And that prevents electrons from going through, right? Because now it's become a layer.

So that's how the cells would, one of the ways the cells would deteriorate is that they actually it starts over time, a little bit of it from the aqueous solution and the electrolyte will actually start to get deposited, right? And that starts to over time deteriorate.

[Speaker 4] (2:13:19 - 2:13:21)

So why is it called lithium ion battery?

[Speaker 3] (2:13:22 - 2:13:23)

Because the lithium ions are sort of...

[Speaker 4] (2:13:23 - 2:13:25)

So they are still the one...

[Speaker 3] (2:13:25 - 2:13:27)

But they're not getting consumed, right? They're staying inside the cell.

[Speaker 4] (2:13:27 - 2:13:29)

Because you're able to reverse the charges.

[Speaker 3] (2:13:29 - 2:13:29)

Yeah, obviously.

[Speaker 1] (2:13:30 - 2:13:45)

Okay, so we understood broadly what a cell is. Now, when people say, like I was talking to that log nine, a bunch of battery companies, which I realized later are not making cells. Predominantly packaging cells into making them batteries.

[Speaker 2] (2:13:45 - 2:13:45)

Right.

[Speaker 1] (2:13:46 - 2:14:00)

How does one battery company compare to another? Why is one better, another worse? What goes into the process of packaging cells into a battery?

You mean in battery manufacturing? Would you call it manufacturing?

[Speaker 3] (2:14:00 - 2:15:12)

Pack manufacturing. Right? No, packing can mean something else.

Packing could happen with a built battery pack also, technically. So battery pack assembly is the right description for it. You can have a ridiculously high amount of difference between companies building battery packs using the same cells.

Who all are doing it in India and what are they doing differently? Mostly it's OEMs doing for their own internal purposes. Explain OEMs. OEMs are basically vehicle manufacturing companies. Original equipment manufacturers. So all of us build our own battery packs typically. Like all the key OEMs in India now build their own battery packs.

We build our battery packs. We don't build batteries, we build our battery packs. And we definitely don't build cells today.

Obviously there are companies like Chetans who are building battery packs because it's a slightly different business. But the differences come from three possible layers. And I think you did talk about it.

One is basically the thermal management. Thermal management is actually, in my opinion possibly the biggest single differentiation between all quality companies. There is actually a level below that.

Just the quality of assembly. It can be very useless. You can just weld the cells wrongly.

[Speaker 1] (2:15:13 - 2:15:19)

Is that only because we are more tropical in nature than China? What is thermal management? Is it because we are hotter?

[Speaker 3] (2:15:19 - 2:16:36)

Let me just describe how actually painfully complex battery pack manufacturing is, assembly is. In a typical, forget the cars. Cars are too many.

Like even in a scooter. In a typical scooter you will have about let's say 200 cells. You will have roughly about 800 welds.

800 welds that will happen for these 200 cells. Two welds on each side. If any of those welds is wrong, that means that cell, that particular cell will lose life very quickly.

Like literally it could lose life within a few weeks. Maybe sometimes months but usually within a few weeks the pack can go off. So you got to get all your 800 welds correctly.

Getting 800 welds correctly for most companies implies a higher level of manufacturing excellence than the famous Six Sigma. Six Sigma is 3 failures per million. Here if you do 3 failures per million welds, that means I think 1 in 1000 or 1 in 2000 battery packs are guaranteed to fail.

Which is a pretty crazy number for a pack manufacturing, for like a OEM. Because the risk of a pack failing in the field is just crazy high. So it is a damn hard job just to get the welding right.

[Speaker 1] (2:16:37 - 2:16:41)

How much of your resources are going into building battery packs out of cells?

[Speaker 3] (2:16:42 - 2:16:53)

Between battery team, battery, BMS and manufacturing engineering, my guess is that somewhere in the range of 60 to 100 people. About 100 people.

[Speaker 1] (2:16:54 - 2:16:58)

And who all are doing it well in India? Battery pack assembly?

[Speaker 3] (2:16:58 - 2:17:07)

I think most OEMs like let's say us, TVS, Bajaj, Ola, Tata.

[Speaker 1] (2:17:07 - 2:17:10)

Is that the big differentiation that everybody is fighting over?

[Speaker 3] (2:17:10 - 2:18:09)

So I think all famous auto companies that you know of right now, they are all building reasonable okay packs. But between us also, you will see significant battery life differences that are not going to be understood yet. So for example, our oldest battery packs are now 5 years in the field.

And they are still at 90% plus health. Which means we are now internally starting to project 8 to 10 year battery life. We know in the industry that I think most of our competitors will also get to 4-5 year battery life.

But 8 to 10 is going to be very difficult for most OEMs. It's a difference that will emerge over the course of a decade. To understand the consumer. That okay, Ather's battery pack, like how Honda's engine became famous over a 10-20 year period.

Perfect rhythm, like no sound, amazing quality. But not visible in the first year or two. But it's a refined difference that will not be understood today.

Most OEMs are building enough sensible battery packs that it won't catch fire for 3-4-5 years. There shouldn't be much difference.

[Speaker 1] (2:18:10 - 2:18:14)

So range is dependent on how many cells are in the battery pack?

[Speaker 3] (2:18:14 - 2:18:14)

Yeah.

[Speaker 1] (2:18:15 - 2:18:17)

That determines range of the vehicle.

[Speaker 3] (2:18:18 - 2:18:19)

So, now coming to the thermal one.

[Speaker 1] (2:18:19 - 2:18:23)

And does it also, like let's say, how much does an Ather scooter weigh?

[Speaker 3] (2:18:25 - 2:18:27)

About 102-106 kgs I think.

[Speaker 1] (2:18:27 - 2:18:28)

And what is the weight of the battery in that?

[Speaker 3] (2:18:28 - 2:18:31)

About 18-20 kgs.

[Speaker 1] (2:18:31 - 2:18:36)

So if you add 50% more cells to your battery pack, your range will go up 50%? Yeah.

[Speaker 3] (2:18:37 - 2:19:21)

Possibly even more for some efficiency reasons, minor. So range is not really... It's not technology.

It's cost. Range is just cost. Range is a product spec decision.

How much range do you want? Obviously, it gets mis-sold as like, you know, somebody's cracked this amazing battery that will do 1000 km in a car. Sir, he has installed two batteries.

He hasn't cracked any amazing technology. So range is generally not a technology problem. Where technology comes in, basically it's thermal management.

What is the difference in thermal management? For example, right now we are the only player where we don't put any restriction on fast charging. So we tell our customers, if you want to fast charge practically every day, be my guest.

Right? Do it.

[Speaker 1] (2:19:22 - 2:19:24)

Otherwise, what is the problem with fast charging?

[Speaker 3] (2:19:24 - 2:19:35)

Most companies, like for example, I have a Hyundai. They recommend, I think, 1 in 10 charges to be a fast charge, I think. I think most companies recommend, don't fast charge if it's a very hot day.

[Speaker 1] (2:19:36 - 2:19:38)

Because it reduces the potential in the cells?

[Speaker 3] (2:19:38 - 2:20:02)

Because your cells heat up. Every time you pass current through your battery pack, current flows

through your cells and the cells heat up in the process. Now, when you turn on the vehicle and the current has to flow through the cells again to go to your motor, the cell again heats up.

In both processes, it heats up. It's like, think of a pipe. If you're like, just basically rubbing the pipe really fast, rubbing the table really fast, it'll heat up.

That's exactly what's happening inside the cell.

[Speaker 1] (2:20:02 - 2:20:04)

And then how do you cool down the bike once it heats up?

[Speaker 3] (2:20:05 - 2:20:13)

That's thermal management. So you can do it multiple ways. In cars, typically it's active, which means there's a fan or there will be a liquid, which will pass through the battery pack.

[Speaker 1] (2:20:13 - 2:20:15)

When bikes have liquid-cooled air cooled?

[Speaker 3] (2:20:16 - 2:21:06)

None of us in India does liquid cooling for two-wheeler. It's a crazy cost structure. It doesn't make sense today.

So, all of us do different kind of passive cooling, which means we'll route air, like when the vehicle is moving, we'll route air through around the battery pack. We will put very interesting materials inside the battery pack that'll take the heat out. There are these materials called phase-change materials.

What happens in them? Basically, if the cell heats, let's say, 50 degrees Celsius, any temperature, any heat above 50 degrees, this material will pull it. And this material will basically become a jelly as it takes the heat out.

So, 50 to a fair degree of temperature, it'll just keep changing. Literally, its phase changes. And then, as it gets time to cool down, it'll again harden up.

So, you've got these materials. You've got a lot of epoxy. We use a lot of aluminium.

Everywhere. Like everywhere.

[Speaker 1] (2:21:06 - 2:21:19)

I might sound really stupid asking this, but I read somewhere that when you stop at a traffic light, that braking kinetic energy is also converted into electrical.

[Speaker 2] (2:21:20 - 2:21:22)

Does it happen in a...? Most regenerative braking, yes.

[Speaker 1] (2:21:22 - 2:21:24)

How does that...? That's very interesting. How does that work?

[Speaker 2] (2:21:25 - 2:21:38)

So, the motor just becomes a generator, right? So, a motor, you know, run the other way actually generates energy back. So, as you press the brake pedal or you put it to neutral, you can get it back.

[Speaker 5] (2:21:39 - 2:21:39)

Right?

[Speaker 2] (2:21:41 - 2:21:47)

In cars, driving, you get probably 18 to 20 percent. That's very high.

[Speaker 5] (2:21:48 - 2:21:48)

Yeah.

[Speaker 2] (2:21:49 - 2:22:00)

I mean, I'm telling you after millions of kilometers of data from the vehicles we have done. 80 to 20 percent. Most bikes...

[Speaker 1] (2:22:00 - 2:22:01)

Is there a factor of the mass?

[Speaker 2] (2:22:01 - 2:22:06)

Well, it's also the way you set the region up. But most bikes in India give you 6 to 10 percent.

[Speaker 3] (2:22:07 - 2:22:08)

Actually, I'll talk about what we do.

[Speaker 2] (2:22:08 - 2:23:12)

And when we've optimized, we've gone up to 18 percent. So, it again depends on the controller, the algorithm, how you integrate it. There's a lot of work that goes behind it and over 15 years of

doing that, we've optimized a lot more on that front.

It's an important piece of it, right? But then also, of course, you think about it when you put a lot of energy, it's charging the battery back, so therefore the batteries heat up. The challenge in scooters in cities like Delhi, when it's 45 ambient, even the air going around it is going to be 45.

So, the batteries are harder to cool down in a midday if you're trying to fast charge it or this. So, that's one of the concerns in the winter. So, climates like Bangalore may be okay.

But that's where active cooling becomes more important. So, if we take the battery and put it in a station, for example, the station's air-conditioned, so the batteries come down to 25 degrees. So, even if the ambient's 45, the battery comes down to 25.

[Speaker 1] (2:23:13 - 2:23:17)

How does that affect, say, aether in Delhi versus aether in Bangalore?

[Speaker 2] (2:23:18 - 2:23:59)

Cells generally at 45 to 25 degrees, you could have almost half the life. That means the batteries are at 45. So, it's very important that that it is.

So, cells don't like hot and they don't like hot and fully charged. And what happens in a cell when it's hot? So, when it's hot, the chemicals start to disintegrate a little bit, right?

So, at a higher temperature, like most chemicals, as you increase the temperature, the deterioration is higher, right? So, a high charge rate and high temperature has a multiplying effect because when there's so much energy there, then there's a lot more corrosion happening internally, right?

[Speaker 1] (2:23:59 - 2:24:06)

So, if I'm buying a 5-year-old electric scooter from Delhi versus Bangalore, I should pay half the price for the Delhi one versus Bangalore?

[Speaker 3] (2:24:06 - 2:26:27)

Half the price. While all of that is correct, and obviously, cells degrade. Like, there are literally these cracks that kind of, I forget the technical term, literally cracks that form that increase the resistance.

More the resistance, harder it is for current to travel and your energy drops. But, what is happening right now is, for the last 5-6 years, and cell manufacturers don't like to claim this because they don't like the liability that may potentially come with it, but most good cell manufacturers have actually pushed the life pretty high. But they don't want to basically claim it,

right?

Because they will be held responsible. So, what we are realizing is, even in a place like Delhi, at the end of 5 years, cells are still probably good at 85-87%. Battery packs, sorry, not cells, battery packs.

In Bangalore, that number might be 90. Now, the difference would be if you run the vehicle a lot, right? Like, I'm sure they will notice, there will be a battery, well, active cooling might make a difference.

What is active cooling? Basically, pass the liquid and keep the temperature low. If you don't do that, and you run, let's say if you run even our scooter, a 100 kilometers or 150 kilometers every day, let's say 30,000 kilometers, 35,000 kilometers in a year, which is pretty crazy, no personal consumer ever will do it, you will notice that the scooter in Bangalore is lasting longer than the scooter in Delhi.

Except that you will likely hit basic life fuel hit. Right now, what's happening is, most, 99% of our consumers, the batteries will literally expire. The calendar aging, they will just literally expire, then they will hit usage.

So, in some sense, if you do enough of a good work with thermal management, whether you are in Bangalore, whether you are in Delhi, whether you fast charge every day or you fast charge once in a week, you ride with three people, you ride with a single person, you ride at 90, you ride at 30 very patiently, you charge sensibly, you never take it to 100%, or you always take it to 100%. All of it is making a limited delta. Similarly, everyone's battery will last for 8 years, some for 7, some for 8.

So, it's not making enough of a difference now.

[Speaker 1] (2:26:27 - 2:27:00)

And what is the story? You often see like these viral dramatic memes of lithium is in Chile and cobalt is in Congo and China controls both the mines. What memes do you read, man?

I say, they come a lot. Don't you see them? So many memes about Congo and Chile and talk about how China controls all refining capacity and we'll be dependent on them forever.

I think something has been discovered in India also now. In Kashmir? But they say there's no refining capacity so even that has to go to China.

[Speaker 3] (2:27:01 - 2:27:02)

It won't take more than a few years.

[Speaker 2] (2:27:02 - 2:28:04)

Just like, you know, there's an area where, you know, lithium, in a way, is the next oil, right? So, I think the government's also thinking what are the strategic partnerships you create? Like, it's created from an oil energy point of view, right?

So, there's already conversations with South American countries, Australia and stuff to say, okay, let's have these partnerships to do it. And that has to be done if certain raw materials and scarcity, right? But also what will happen, Nikhil, in 10 years, right?

Recycling would play a very large role in a market like India because these batteries, once they're off, you know, today the cell phone batteries and everything were very small and they were very different formats. But, you know, as one of these car batteries may be equivalent to 5,000 cell phone batteries. So, any recycling is there.

And today recycling costs are actually earlier you used to pay to recycle, now you can make money out of it. If anything can make money in a market like India where you sell paper and recycle it.

[Speaker 1] (2:28:04 - 2:28:12)

Make money out of it how? It means when you recycle it, you can sell back more than the cost of recycling. Oh, damn!

[Speaker 3] (2:28:13 - 2:28:30)

For example, just the cost of material, just the material alone in a typical 4 kilowatt battery pack is about $240 today, right? About $60 a kilowatt hour. This is the raw material inside.

Literally the cost of material, not like the assembled, processed stuff.

[Speaker 1] (2:28:30 - 2:28:32)

Cost of recycling the material to get it back?

[Speaker 3] (2:28:33 - 2:29:16)

Possibly 1000-2000 rupees, 10-15 dollars, 20 dollars. So, if you like, one of the places where people should go after in India, guaranteed is recycling. Granted, you will not get that entire $240 yourselves because enough people in the chain who will take that money away from you.

But the delta is so large that even if you make like 2-3 thousand rupees in a pack, it's huge. And the national policy is almost going in the direction of lithium once in the country should never exit the country. Once if you import lithium with so much pain and forex, bloody make sure it never, please make sure it never leaves the country accidentally.

[Speaker 1] (2:29:16 - 2:29:47)

Is that a viable business? I was going to come to this question. From all that you spoke about, you started this business in 2013-14 with a friend of yours with less capital.

If I am a young entrepreneur today, wanting to start something here, where do you see the opportunity? Is it in cell recycling? Is it in trying to build?

I'm going to come back to the bike itself. You said 20 kilos out of 104 is the battery. Then we'll go to the chassis and the other parts.

But where do you see the opportunity?

[Speaker 3] (2:29:48 - 2:30:31)

So I think recycling of cells is a very big opportunity. I think there are captive requirements. Like for example, if you came to us today and you said, Hey, I will run basically a contract shop for you.

You give us all your reject battery packs. I will strip them down. I'll convert into raw materials, which you can sell to your cell supplier back.

Right? It essentially fixes my pricing, right? If the price of lithium increases, I have to pay more for a production for a month.

Right? If I have enough cells coming back, then I can just basically remove that fluctuation out. So what do I need to start this business?

There is enough tech available. You today will probably need a JV immediately if you want to fast track.

[Speaker 5] (2:30:31 - 2:30:31)

With?

[Speaker 3] (2:30:32 - 2:30:40)

With some company in Europe or probably in China or Korea who will have recycling tech. Or you can spend a couple of years and just figure shit out.

[Speaker 2] (2:30:40 - 2:30:41)

But there are companies in India.

[Speaker 3] (2:30:41 - 2:30:42)

There are good companies in India also.

[Speaker 2] (2:30:43 - 2:30:45)

Who are doing recycling and are setting up shops even outside India.

[Speaker 3] (2:30:46 - 2:31:18)

So you set up recycling. I think it is such a stable business. So much battery is going to be produced in India.

250 million vehicles going to go to 350 million vehicles. And annually out of that 30 million is going to come back. And 10-15 years later, a very large chunk of that will be electric.

You got almost a guaranteed business. Right? Sure, you may not make like a 50% margin business.

But you don't have to worry about demand at all. What if I want to start an electric scooter? Then you will need something disruptive.

[Speaker 1] (2:31:19 - 2:31:22)

Like? What do you think the next disruptive thing there is?

[Speaker 3] (2:31:24 - 2:31:45)

Give away my road map. See, so I think the market has changed from the time we started. When we started the disruption was the spec.

Us saying that when we started the best electric scooter in the market did 25, maybe 30 35 kilometers per hour. Top speed. 800 watts.

Really, it was a mixer motor.

[Speaker 1] (2:31:45 - 2:31:51)

This also you didn't answer. We said double the number of cells, double the number of amount of range.

[Speaker 3] (2:31:51 - 2:32:20)

Correct. What determines the speed of the scooter? Motor.

Motor power. And the motor power is determined by the battery size in some cases. So bigger the battery it can supply more current.

And if you want more speed, you need a bigger motor. Rule of thumb, you want to go from let's say every 10% speed increase increases the motor power by roughly about 30%. Something like that roughly.

So if you have a bigger motor, you'll go faster.

[Speaker 1] (2:32:20 - 2:32:26)

So are these the main parts? There's a motor, there's a battery pack, there's a chassis, there's tyres, whatever like body.

[Speaker 3] (2:32:26 - 2:32:30)

There's a charger. There is a motor controller, another piece of electronics.

[Speaker 1] (2:32:30 - 2:32:33)

Is that what determines how much current goes into the motors?

[Speaker 3] (2:32:33 - 2:32:34)

Yeah. Motor control is quite important.

[Speaker 1] (2:32:34 - 2:32:35)

And that determines the top speed.

[Speaker 3] (2:32:35 - 2:32:41)

Actually a lot of the IP today for companies sits inside the motor controller. Motor itself has limited IP these days.

[Speaker 1] (2:32:41 - 2:32:44)

Can you break that down a little bit? Explain how a motor controller works.

[Speaker 3] (2:32:45 - 2:32:53)

To run a motor, you need somebody to control, somebody to send signals and control the power that it gets.

[Speaker 1] (2:32:54 - 2:33:00)

In my mind, I'm thinking IC vehicle. I'm thinking there is a spark plug. As is expanding.

[Speaker 2] (2:33:00 - 2:33:02)

Pistons. Yeah, but think of it very simply.

[Speaker 1] (2:33:03 - 2:33:05)

You have a regulator for your fan in your house.

[Speaker 2] (2:33:06 - 2:33:13)

Right? For your fan. You kind of do one speed, two speed, three speed, four speed.

The motor controller does that in a very sophisticated manner.

[Speaker 3] (2:33:14 - 2:33:16)

What is the uniqueness?

[Speaker 1] (2:33:18 - 2:33:27)

So what is the big unique, like what is the differentiation between your motor controller and Ola's motor controller? I'm just taking you guys because you're the most popular.

[Speaker 3] (2:33:27 - 2:35:08)

So I don't, I haven't exactly studied the motor controller in detail. But I can give one feature example. Recently that both of us did at the very same time.

But just about 10 months ago. Both of us implemented a feature called auto hold. Cars have auto hold.

Hill hold. Right? So for example, if you're on a slope, like coming out of a mall parking lot, let's say, you will engage auto hold, which means you will never accidentally slide back.

So if you put the throttle on, you will go ahead. If you leave the throttle, you won't go back by mistake. Simple feature.

Two wheelers don't have it. But in electric, you can build it. Because you've got a very sophisticated motor controller.

So it can ensure your motor only turns forward. Does not accidentally go into reverse even if there is force coming on it. And it's a lot of sophisticated control to ensure how to manage that load, and ensure you always move forward.

But the way both of us implemented it, pretty substantial difference. It fundamentally sounds the same feature. But the way we implemented and the way Ola implemented, I think Ola's same capability would last a much shorter time.

So if you engage this hill hold feature on a steep slope, after a minute or so, the motor would be like, okay, now I can't support this. It will start rolling back. The way we implemented it, you could just stay like that for 10 minutes.

So there is something very different about how we did the algorithms, about how we managed the heat coming out of the motor cooling, the motor heating up. A lot of very different stuff. We also changed just the experience.

Like you don't have to tap this, press this button, do this. In fact, cars have it more complicated. We said, if you press the brakes and you release them, you're automatically in hill hold.

You won't go back or forward.

[Speaker 1] (2:35:09 - 2:35:24)

I have to buy a scooter, okay? Both of you cost similar amounts. Maybe 10,000 bucks here and there.

So how do I pick, should I buy Ola or should I buy Ather? This is a very relevant question for everyone watching this show, which will be millions of people.

[Speaker 3] (2:35:25 - 2:35:29)

My pitch is you should buy us. Because you probably understand what we have done.

[Speaker 1] (2:35:31 - 2:35:32)

What have you done?

[Speaker 3] (2:35:33 - 2:36:02)

Before that, you asked a question at the start and I'm doing my company injustice if I don't answer the second part of that question, which is, you asked me what did Ola do well and what is, but I didn't answer what is Ather doing well. I would say what we're doing well is we have better word of mouth. So we may not have marketed the product well.

So not enough people may buy it today. But the ones who are buying it have a better word of mouth for us than many of our competitors. Because?

Because of how the product is built. It's overall quality, the overall experience that you get.

[Speaker 1] (2:36:02 - 2:36:09)

Why are large scooter companies like Ola getting that PLI benefit and Ather is not?

[Speaker 3] (2:36:11 - 2:37:01)

Because PLI was just written incorrectly. And after the mistake was made, everybody's still struggling, who will correct it? Because correcting a government written document is just, who wants to build that cat, right?

Who wants to be the guy who changed the document? So everybody's super scared of touching it. Though almost everybody agrees unanimously that's a mistake.

So the definition was either you have a revenue of 10 crores, which no EV company has ever made. Which means only listed large auto players who actually have very little interest in actually doing electric could apply for PLI. Or the only other way you can apply for PLI is if you actually are not manufacturing electric vehicles.

But if you are a company manufacturing electric vehicles, making actual investments, committed 100% only to doing electric, you won't apply for PLI.

[Speaker 1] (2:37:02 - 2:37:05)

So if I start a new scooter company today, I won't get the PLI?

[Speaker 3] (2:37:06 - 2:37:38)

No chance. And it's not even like there's a second window. A window opens up every year that you can apply for PLI.

So it's a pretty weird mistake that's happened. It needs to be corrected. But that's the thing.

Somebody just needs to find the courage to acknowledge that this is a mistake and it needs correction. And it's not just us. We are probably the most permanent one in the E2Villa industry.

But as you said, if you were to start an E2Villa company today, you're kind of screwed. You can't apply for PLI. Even if you say, I'll invest 2000 crores into this, give me PLI.

You still can't.

[Speaker 1] (2:37:39 - 2:37:54)

Okay, so let's go break down the cost of the scooter by weight, just to make it interesting. 104 kilos, 20 batteries. Chassis, how much?

Start from the biggest weight. Chassis is the largest.

[Speaker 2] (2:37:54 - 2:38:26)

Chassis really are going by weight. Plastics go by weight. Tires go by weight.

Right? A lot of those core components which are there in any scooter will go by weight. The

differentiation is your motors, your controllers, your battery packs, your clusters, your telematics, right?

Those are areas that unfortunately the weight equation is not necessarily there. It's about, I mean, just take your iPhone. And I'm just saying, how many grams is that?

[Speaker 5] (2:38:26 - 2:38:27)

Why does it cost?

[Speaker 2] (2:38:27 - 2:38:30)

I'm just trying to get that piece as an example.

[Speaker 1] (2:38:31 - 2:38:42)

I'm not asking for weight in the sense that he'll say 100 kilos and I'll be like, you're talking too much. So cost-wise, battery cost, how much?

[Speaker 3] (2:38:43 - 2:38:47)

Batteries, I can't give the exact numbers, but in percentage, generally about 35% is a good estimate.

[Speaker 1] (2:38:48 - 2:38:53)

35% of 1,65,000. Close enough. Okay.

[Speaker 3] (2:38:54 - 2:38:56)

That's not my cost, right?

[Speaker 1] (2:38:56 - 2:38:57)

My cost is different.

[Speaker 3] (2:38:58 - 2:39:09)

Okay, next biggest component. Next biggest component is in our case, the motor plus controller.

[Speaker 1] (2:39:10 - 2:39:12)

Not the motor part we discussed earlier.

[Speaker 3] (2:39:13 - 2:39:41)

Oh wait, motor and the electronic controller that goes together. How much would this be? You

can take this as another maybe 10%.

Then you can take the chassis. Actually, chassis is the second component. Chassis is, I think, I would say about 15%.

So the frame and all the adjoining components. Then you've got, in our case, a touchscreen dashboard. So that's another 10%.

[Speaker 1] (2:39:41 - 2:39:42)

The 70% then?

[Speaker 3] (2:39:42 - 2:39:46)

Then you've got a charger.

[Speaker 1] (2:39:47 - 2:39:49)

Charger, external charger.

[Speaker 3] (2:39:49 - 2:40:09)

Charger and some, a lot of peripherals. That's another 10%. Then you can add everything aesthetic.

The lights, plastic, body panels, all that. I'm simplifying it, you can take 10% more. The rest is basic accessories.

10% is left. Basic stuff. There's a lot of other electronics on the vehicle.

[Speaker 1] (2:40:09 - 2:40:12)

How much are you selling the scooter for now?

[Speaker 3] (2:40:13 - 2:40:20)

Between 1 lakh 30, 1 lakh 25 actually. 1 lakh 25 to 1 lakh 65.

[Speaker 1] (2:40:21 - 2:40:34)

1 lakh 25 to 1 lakh 65. The subsidy per scooter in Fame 2 used to be about 40,000 rupees a scooter? 60.

Now it is 40?

[Speaker 3] (2:40:35 - 2:40:36)

20, 22.

[Speaker 1] (2:40:36 - 2:40:40)

So your cost of bike from 125 to 165 is going to?

[Speaker 3] (2:40:40 - 2:40:43)

That's our selling price. My cost is different.

[Speaker 1] (2:40:43 - 2:40:44)

What is your cost?

[Speaker 3] (2:40:45 - 2:40:46)

Ballpark?

[Speaker 1] (2:40:47 - 2:40:53)

You can see the margins in numbers, anybody will find out. Chetan, what is it?

[Speaker 2] (2:40:56 - 2:41:03)

155. To that you have to add the subsidy, right? Because the subsidy is coming to the customer, right?

[Speaker 3] (2:41:03 - 2:41:07)

No, to them. The 155 is the customer's total price, which means 20,000.

[Speaker 2] (2:41:08 - 2:41:14)

So in reality, the selling price is this plus 20 in both the cases.

[Speaker 3] (2:41:14 - 2:41:18)

But you remove 5% GST. But this is crazy.

[Speaker 1] (2:41:19 - 2:41:27)

If on a 1 lakh 65,000 scooter, you get 60,000 on top of what you sell for, right?

[Speaker 3] (2:41:27 - 2:41:27)

Used to.

[Speaker 1] (2:41:28 - 2:41:51)

So if he was getting 2 lakh No, you have to account for GST, so 2 lakh roughly. Without GST, if he was getting paid 2 lakh 25,000 in May 23, now he is getting paid 1 lakh 65,000 when he sells the same scooter. Okay, I am going to make an assumption that you are spending 1 lakh 75,000 on a scooter you are selling for 1 lakh 65,000.

[Speaker 3] (2:41:51 - 2:41:58)

Come on. Roundabout. No, it's lesser.

[Speaker 1] (2:41:58 - 2:42:01)

Okay, let's say 1 lakh 55,000 then.

[Speaker 3] (2:42:02 - 2:42:04)

You take 1 lakh 50,000.

[Speaker 1] (2:42:07 - 2:42:16)

So how do you work in an ecosystem your cost for the same commodities have not gone down significantly, right?

[Speaker 3] (2:42:16 - 2:42:23)

Sales has come down a bit, a fair bit. But that's about maybe 10,000 a piece since the start of this year. Commodities haven't that much.

[Speaker 1] (2:42:26 - 2:42:28)

Sales have come down by what percentage?

[Speaker 3] (2:42:30 - 2:42:31)

25%.

[Speaker 1] (2:42:32 - 2:43:01)

But still, this is a significant hit to your margin, right? So this is another question for anybody brave enough to take up the job of building an EV today. I have some friends who are trying it.

There's Ultraviolet, there's some guy trying to build a car called Praveig. These are all Bangalore boys, like old friends of mine. How do they plan for action from the government?

I know Fame3 is coming. Is there a prediction you can make around Fame3 to help someone new start?

[Speaker 3] (2:43:01 - 2:44:02)

No, actually Kanan, that's one of the reasons why it's super difficult right now in this environment to actually be a startup trying to enter breakthrough space. Because you just have no basis. If you say I want to build a product that I want to price at 1,40,000.

But that's for a Rs. 22,000 subsidy. What if this goes down to zero?

Suddenly you may discover you don't have a margin. In which case your burn goes up massively. Right?

So your entry point to building an e2 wheeler in the market today, your table stakes of just entering this business, in my opinion is about 100 million. If you're bringing 100 million, you can enter the game and then you'll have to raise a lot more. Right?

The only way out of it and that also requires a ton of money is a phenomenal amount of cost reduction. Which we've been doing, everybody in the industry is doing but it's super expensive. I'll give you a sense.

I have a 1000 member R&D and we do very heavy analytics on this. You have a 1000 member R&D? Yes, that's the world's largest R&D for electric two wheelers.

And we do very heavy analytics on this.

[Speaker 1] (2:44:02 - 2:44:11)

And this is not a thing, right? e2 wheelers are only big in India, right? In Southeast Asia, it's 3 wheelers, 4 wheelers.

I remember reading taxis converted first in China.

[Speaker 2] (2:44:12 - 2:44:19)

Yeah, I think that the China market is a very different organization. It's highly commoditized.

[Speaker 3] (2:44:20 - 2:44:21)

Super cottage industry.

[Speaker 2] (2:44:21 - 2:45:24)

Yeah, so for example, there's companies that's making motors and they specialize at very high volume. They make 2 million motors. So the pricing is very low.

They have type A, type B, type C motor. So if you're a two wheeler manufacturer, you just go and say, I want that motor. I want that controller.

Very inexpensive made in very high volume, right? It's also replacing markets easy. I have standard wheels, I have standard tires, right?

So a lot of the battery packs are coming from you know, there's 1500 battery pack manufacturers in China. So you can just almost commoditize it and look at it. So when everything is commoditized, then you're just doing your frame and body.

They're getting much simpler solutions. They're not doing high-end scooters, they're doing basic scooters. For that, the investments in those are not very high.

So they're probably for, you know, a few million dollars, you can start a 2 wheeler company, right? And be and survive in that space. And it makes sense.

But that's very different from you know, where India is, or where the premium market is.

[Speaker 3] (2:45:24 - 2:46:13)

That could only have happened in China. See, because in China, the government force created supply and demand for electric 2 wheelers. Correct.

So the way all these manufacturers came up, you force created so many suppliers. Like motor manufacturers, I remember when I started Ather, and I read this ADB report, there were 400 manufacturers building electric motors for electric scooters in China. And I don't know how many hundred manufacturing companies.

It's a super fragmented market. But, while the entry barrier is small, the reward is also small. Because no company makes enough margin to really build a large valuation or a large market cap.

There's just not enough value captured by any real player. Everybody is essentially a glorified commodity manufacturer. India has gone down a very different path.

India has gone down, I would say, the more the US path.

[Speaker 2] (2:46:13 - 2:47:13)

You know Nikhil, what Tarun was saying, at the top level on a policy, the government is very aligned. But there's this gap between the vision and the implementation. So one example, what we talked about is a PLI scheme.

The same thing is coming for the ACC, right? It's only given to a few players, very big players, which means some of them will be very successful, but that doesn't build the entire industry up. How do you create a sort of level playing field on policy?

So when you look at the GST, electric vehicles at 5% GST. But a battery pack, batteries are at

18%. So it's also hard for a lot of companies only making electric vehicles where they'll have an inverted duty structure. Because the input will over time be more than your output.

That happens to us.

[Speaker 5] (2:47:13 - 2:47:13)

Right.

[Speaker 3] (2:47:13 - 2:47:15)

We'll earn tons of money every month.

[Speaker 2] (2:47:15 - 2:49:36)

But that will not be an issue for a company that also has an IC engine business. Because there your GST is 28%. So now, if you're an EV specific company, you're a disadvantage in a company that is EV and IC.

If you take that again to swapping, because the battery is at 18%, and now you separate them out, suddenly the battery is at 18% but the vehicle is at 5%. Now when a customer goes to swap, this auto driver is paying 18% tax. Crazy.

On going green. So that should really be 5%. So there's a gap if the government say we want EVs to be at 5%, then the inverted duty structure should disappear, and services related to that for charging and swapping should come at 5%.

Subsidies. Let's say tomorrow, I mean today we have one of our OEMs, a PR show. In their showroom, there's a vehicle with a fixed battery and one with swappable.

Both have identical vehicles, you can't tell the difference between them. You buy the one with fixed, the customer subsidy comes through. To buy the one with swappable, the subsidy doesn't come through.

The customer doesn't get it. Like he's today passing the subsidy through. So these anomalies are really challenging where they've not created a clear technology agnostic plan.

That's why I think there's a big gap between, like he just said, and once this is in place, everyone's going round and round. There has to be a far more holistic thinking that how policies with technologies can be agnostic. Things change over time.

You can't say some things like this, but if you create fundamental ways of thinking, then all of them will flourish and come through. But today it's highly disadvantageous for companies in this way. And that's not the right thing we should be doing.

We should actually, like you said, promoting younger startups, allowing know that whichever way you go, the customer still gets a subsidy the same way. All policies are similar. And give benefit, if anything, to startups.

That's how US and everything has. If I look at the right time when Tesla was almost going off, right, they got $500 million from VC backed by $500 million from the government. That billion dollars at that right time made Tesla what it was, right?

[Speaker 1] (2:49:37 - 2:51:18)

That's what, I mean, we're doing that to companies today and supporting it, a very different situation. But cost saving per vehicle, I wanted to ask you as well. How much does it cost you to ferry a passenger for one kilometre on your vehicle versus a IC car?

I can talk for the kilometre in our case is about $22, $18 to $20. Depends on what product you take. You take a point-to-point ride or rental.

It's in the range of about $18 to $22. Is the our brand promise. What is your cost?

See our cost is lower but our utilisation has to go up to make for that. At current utilisation? I mean, look, if I were to say a car should make about, to be self-sufficient, to break even on all, should make about Rs 3,200 per day to cover all my costs.

The car has to earn me Rs 3,200 a day. It includes driver cost, lease of the car that I'm paying on a daily basis. How much does the driver get paid on average?

About Rs 800 on the lower side, Rs 1,000 on the higher side. So driver takes that. The car EMI or the lease on daily basis, monthly divided by number of days is about Rs 800.

Parking, charging, all the paraphernalia is about Rs 580-600. Then the miscellaneous is your toll and your other costs. That's about Rs 400-500.

It all comes down to about Rs 3,000-3,200. Cost has been reducing drastically to Rs 3,500. Now it came down to about Rs 3,000.

I have heard that in scooters, if let's say a Honda Activa costs Rs 90,000 per scooter, their bikes cost Rs 1,500,000.

[Speaker 3] (2:51:19 - 2:51:22)

No, no, not apples to apples. Apples to apples would be Rs 120,000.

[Speaker 1] (2:51:24 - 2:52:26)

So what I've heard is if somebody were to ride the scooter for 40 km a day, they save Rs 35,000

a year in terms of cost of energy in fedding that vehicle. Accurate? Yes, Rs 25,000.

Is there a number like that for your cars? If you run a lakh kilometres at Rs 25 paise, Rs 50 paise. Lakh kilometres would be how much per day?

You would be driving 250 km because at 200 km you're doing about 6,000 km a month, 72,000 km a year. I'm just giving a ballpark number. Okay, let's come down to 72,000 km only.

This will also help me make up my mind if I should buy a Tata IC or electric. You should definitely buy an electric. I'll give you a layman number.

If you run 3,000 km a month on an MGE, your bill will be Rs 3,000. See, for a simple man. And in IC?

It will be much more. I mean, your cost will be about Rs 24,000 per km. Depends.

[Speaker 3] (2:52:26 - 2:52:27)

Depends on what car you're using.

[Speaker 1] (2:52:27 - 2:52:29)

Rs 30,000 versus Rs 3,000?

[Speaker 3] (2:52:29 - 2:52:52)

Actually, Nikhil, for you, it'll never make the ROI. See, unless you buy a low-cost electric car, the problem is the asset cost of an electric car is so high that the km saving is only worth it if you buy a car in the 10-20 lakh car range. If you buy like a 80 lakh, 40 lakh, even a 40 lakh electric car, you don't have an ROI.

Unless you run it like 100 km a day.

[Speaker 1] (2:52:52 - 2:53:43)

So, what I see a lot of people now, they're shifting to electric because they're saving on the fuel cost. They're not looking at the EMI of the car because that's anyways, in EMI they're paying over 7 years. That's a massive saving.

So, between a Skoda Rapid and an MG electric, the difference is Rs 21,000. You have Rs 6 per unit kWh for the BST meter, say 50 kWh battery, it's about Rs 300. The range of the car is 350 but I'm saying 300 km.

300 divided by 300, 1 rupee per km. You remember old school people used to sell cell phones like AT&T? You'd buy like a Blackberry back in the day.

Then AT&T started this thing that don't buy the cell phone, pay me a monthly fee every month. Wouldn't that be a natural next step for you guys because you're saving the monthly cost? Why sell a scooter if you can say pay me 5000 bucks a month and use it?

[Speaker 2] (2:53:44 - 2:53:53)

That's what we do in our mobility service. We give you a vehicle, energy, data, services, all integrated with maintenance and all. You run your business.

[Speaker 3] (2:53:54 - 2:54:43)

So, we give that full solution. Wouldn't that make sense to you? We failed when we attempted it.

You attempted it? We attempted a variant of it. So, I do believe in it and I still do and I'm hoping that we learn how to crack it.

But back in 2020, so what happened for us is we were selling our vehicles at 1.2 lakhs back then and there was this moment of reckoning that the money is running out in the startup and loss is too high for every vehicle that we sell. So, I went back to basics, did the math, concluded that if we do everything right from my engineering, we'll bring the cost low down enough that, but even then we can't profitably sell this at 1.2. To make money selling this vehicle, we need to up the price. We need to make the price 1.60 lakhs. So, in one shot, we had to go from 1.20 lakhs to 1.60 lakhs. So, obviously, nobody's really got the **** to announce it very quickly. But I'm convinced.

[Speaker 1] (2:54:44 - 2:54:46)

But maybe that's an opportunity for an entrepreneur.

[Speaker 3] (2:54:47 - 2:55:12)

Buy your bike, start a leasing business. So, while we did make it, announce it 1.60 lakhs, as a way of softening the blow, we said, actually a version of battery swapping, we said you could take a battery on subscription. Okay, from us.

Which basically means the battery is in the vehicle. It will never be removed. Except that, you don't own the battery.

You just pay some monthly fee. So, you can get the vehicle for only 90,000 rupees. Okay, 99,000 rupees.

[Speaker 1] (2:55:13 - 2:55:18)

But why did you think 90,000? You should have been like, freeze, check their credit score, charge 5,000 bucks a month.

[Speaker 3] (2:55:18 - 2:55:57)

So, let me tell you the problems that we ran into. Right? And we said 1500 rupees plan, 1800 rupees plan, lot of complications we came up with.

Enormous, massive community pushback. Because the biggest feedback that we got was that, vehicle is still such a passionate, emotional purchase for me. I need to own this.

So, I'm not gonna pay you a lifetime fee for this. You tell me when does this EMI stop? I mean, if I make an EMI, then it's an EMI.

Like, I'm trying to make a subscription product and make a lot of money out of this. You tell us, 2 years, 3 years, 4 years, when does the EMI stop? So, ultimately, I need to own this battery.

[Speaker 1] (2:55:57 - 2:56:08)

This is a very emotional thing. It might work in another country. So, we couldn't convince them.

I think we would all agree that a transition to EV is happening. Even Suruchi, right? Absolutely.

No? No doubt.

[Speaker 3] (2:56:08 - 2:56:09)

I didn't realise she was in agreement.

[Speaker 4] (2:56:12 - 2:56:23)

Unfortunately, even I, just on our way here, you see so many EV showrooms, right? And I'm just talking about cars here. The revolution is here.

[Speaker 2] (2:56:24 - 2:57:20)

I've never seen everything like this come together. Right? A, you think energy prices have come down and crossed over.

Look at government support. I mean, I'm going back in my sight 20 years ago. We still have challenges.

Tarun and I spoke about it. But they have fame. They have road tax exemptions.

They have a lot of concessions. Governments are for it. Right?

If I look at battery pricing, the way it's come down. If I look at technology innovations, today, the performance you get from these vehicles. If I look at consumers today are willing to look at electric.

10 years ago, they weren't. You at least have it in your area. It's very rare that every piece of it has come together.

Right? I think that's where you have the macro conditions are absolutely perfect today for this industry. So it's just about, and that hasn't happened.

I'm just seeing it in 20 years. You see the pulse, and you look at everything. Everything has to come together to really make it happen.

I think that's where you are today.

[Speaker 1] (2:57:21 - 2:57:50)

I was trying to create discourse. Personally, I'm looking to invest in EV and I've been researching a whole bunch of companies. Not just in EV, but energy transition per se.

Carbon tax. A big part of Tesla's revenue for many years has been coming from carbon credits. Not so much today, but at least in the very early years.

Do you think such a thing will happen in India and that could be the inflection point for EVs?

[Speaker 4] (2:57:50 - 2:58:05)

There is a carbon tax in conversation and they've put out a framework for estimating this tax based on which part of the value chain you're based out of. It's still complicated and we're still trying to get through it and understand what the outcome is.

[Speaker 3] (2:58:07 - 2:58:11)

I'm almost certain a carbon tax is headed our way in India.

[Speaker 1] (2:58:11 - 2:58:15)

Do you think the world will be taxed at the same rate and do you think that would be fair?

[Speaker 3] (2:58:15 - 2:58:31)

The world or India? The world. Because carbon tax only works if everybody's.

I think what's going to happen is we're going to be forced into it because that example that's going to happen to us multiple times and I know that folks like Niti are already annoyed by it. That all the goras are going to tax the heck out of us with carbon.

[Speaker 1] (2:58:33 - 2:58:34)

They've grown on the back of carbon.

[Speaker 3] (2:58:35 - 2:59:30)

So now it's time to tax us. So I know, I do believe, I can already see it at some extent that policy makers can see it coming that we will be taxed first as a country and so that's one factor. It will eventually flow through.

The other thing is the only thing government really can do is like the EV transition, the green energy transition has to be financially supported in some way. It could be a tax incentive, it could be a subsidy, it could be a PLI, it could be whatever, right? Like you're supporting it and the scale has to be built to like a gigantic scale, right?

At some point, like for example in electric, EVs are kind of taking on its own, like without any support also. A fantastic opportunity is to then tax the vehicles that anyways are sort of on a downtrend. Nobody is really like lobbying for them now.

So I think a carbon tax will find its way into everything. Vehicles are the first.

[Speaker 1] (2:59:30 - 3:00:00)

My proposal, BlueSmart's proposal, I was in the CII Mobility Chairperson last year, Chairman for Mobility Service. And my recommendation was, even to CII and to the Delhi government was impose ultra-low emission zone charge. Could be any price you come up with, with your consultant or some advisor.

Charge each CNG, petrol, diesel car, this amount. Collect that money, make a pool, use that for building public charging infrastructure. This is what we both are doing.

Government also comes in, uses that pool to set up charging.

[Speaker 2] (3:00:00 - 3:00:40)

So Delhi already has started this almost 12-13 years ago. They created the Ambient Air Quality Fund, which taxed a certain paisa for every litre of diesel and petrol. And that put it into this Ambient Air Quality Fund, which will only be used for EVs.

The reason Delhi still has a subsidy corpus, and is still able to give international, is from this Ambient Air Quality Fund that was started over 12 years ago. So it already has this, which came in the time of Sheila Dixit, at that time. And that's sustained it.

That's why you have the best city incentives in any place in the country is in Delhi, and it's continuing.

[Speaker 1] (3:00:41 - 3:01:49)

It's gone on for years, because they just have the corpus that's built up. I only hope when a

carbon tax comes, it is not applied per person, but more... On usage.

It should be applied on usage. Because we as Indians use significantly lesser energy even today than the West, right? Yeah.

Okay, last question guys. Tell me one suggestion you would make to a budding entrepreneur who's 20 years old. What should he do in the entirety of the conversation we had?

One low-hanging fruit he should start a business around. I'll start with Puneet. I'm pretty convinced battery recycling is a massive opportunity.

I would not say do charging, because we are doing it. The low-hanging fruit, I mean if the EV adoption is imminent, massive opportunity to build charging. Because we are doing it, so I won't request somebody else to come and compete with us.

So, what we don't want to do is to do battery recycling, so wish you all the very best, you can do that. Big opportunity, go chase your dreams. Waste management, EV waste management.

Yeah, that also could be another related party.

[Speaker 3] (3:01:50 - 3:01:51)

It's in the same recycling space.

[Speaker 4] (3:01:53 - 3:02:37)

I'm genuinely thinking about setting up EV charging at our offices, but with the hopes that eventually the electrons, we produce a lot of hydrogen when we're testing. We have like a demo plant, it produces 10 to 25 kilos, that's a sufficiently large quantity of electricity, as long as we get a good enough fuel cell to convert that to electricity. Maybe you should set up one charging facility in every industrial complex.

I feel like it makes a lot of sense because there is a lot of off-gases that don't always get consumed, testing, downtime, etc. That'd be just enough. There are cars in Jamshedpur that drive people around.

You can't really walk, it's a huge plant. Seems like a wonderful opportunity.

[Speaker 2] (3:02:38 - 3:03:53)

So, if I see the next five to seven years, and I think of entrepreneurs, it's hard to get a lot of money initially, right? What do you do from that perspective that allows you to enter a business and grow it? I think the way energy and mobility is going to shape up, the intersection of that is going to be a data play and an AI play and a lot of analytics coming in.

And this is going to be new for these new businesses, right? So, I do believe there's a large opportunity where a lot of entrepreneurs in India who got that with these policy changes and these markets coming in, right? So, these markets don't exist.

And if they're going to get bigger, you're going to find a way to connect them. And there aren't solutions today. You're doing something, we're doing something, but there's going to be much bigger pieces of that solution.

So, I think that's a good opportunity which is new, which is global, which is data, AI integrated, and probably allows people to look at a particular problem that's very specific to, say, BlueSmart or us or someone else, and solve that and say, in the process of solving this, I'm giving a value and I can get in quickly and make that happen.

[Speaker 1] (3:03:54 - 3:04:19)

I would say build an adapter. I'm not knowledgeable here, so I'm writing like Marvel comics while I'm saying this. Build an adapter for whatever electric vehicle you have, which makes it compatible with what he has, what he has, what you have.

And then map out where all the charging facilities of India are and figure out a way to communicate with them and find the closest one. We'll make it cheaper.

[Speaker 3] (3:04:22 - 3:06:41)

So, two ideas. One, I would say, if you look at Southeast Asia, two-wheeler penetration is very high. If you look at the rest of the world, overall automotive penetration is very high.

So, we are going to sell a lot of vehicles still, and I think personal mobility is not going away, obviously. It's going to be pretty big, especially as Indians become richer. I think for a while, we're going to be in that phase where everybody will have aspirations, you'll want to buy more.

Build a new form factor. I think electric allows that opportunity that ICE did not allow you. ICE kind of bucketed, because of technology, only a few form factors for you, car or two-wheeler.

I think electric allows you the opportunity to, for example, the entire explosion with micromobility, the uniform factor, right, and those vehicles. I think there are many such opportunities. So, if you want to start an OEM company today, don't just go and say, okay, I'll also build a scooter.

It's already like now happening for enough years. That's not disruptive. Go change a new form.

Attempt a new form factor. I think a lot of opportunity. Second one is something we failed at.

I didn't succeed. Definitely didn't succeed at that point. But I think there's a big opportunity, and I agree with you guys, that yes, there are, I think, a lot of customers who want a different form of

ownership with great vehicles, which can be expensive.

Crack that. You can just simplify and say, crack a subscription model, but own the vehicle, not like a daily subscription, right? How do I give you the sense of ownership?

These fancy car companies have that, you know, Mercedes, lease and buyback. Yeah, but somehow, they're not like really flying off, right? So, how do you make this work for a very large chunk of customers suddenly?

I think there's a big opportunity there. And again, electric and specially connected vehicles, amazing enabling platform, right? Connected vehicle can be tracked at all points, you know the usage, you know where to send them for charging.

So, I think the ecosystem is really well set for this kind of a company to be built. I think both opportunities are there. And surprisingly, both, at least the latter, a lot of people thought about it, but nobody's really cracked it.

Every company who's talked about cracking subscription, quickly, quietly became a financing company. Everybody. I think so the opportunity is still pretty much open.

And so is the opportunity for a new form factor.

[Speaker 1] (3:06:42 - 3:07:28)

Okay, last part, at the end of every episode, we do something which is going to help young entrepreneurs or something charitable in nature or funding a company which would have otherwise not got funded. I ask each of you to give away a certain amount of money, whatever you want to commit to. And then we will find a project together, we will find that young, bright mind somewhere who could both use your mentorship, all of you, and also your money as a grant or as funding into his venture.

So, you guys can commit an amount and then we'll reconvene and we'll go through the list of entries and we'll select the final one. Who'd like to start?

[Speaker 3] (3:07:29 - 3:07:33)

So, if it's an investment opportunity, I think I can put 5 lakhs.

[Speaker 1] (3:07:33 - 3:07:34)

Okay, Chetan?

[Speaker 2] (3:07:35 - 3:07:40)

I would also look as investment opportunity. In fact, even given my time, if I find that idea is really

good.

[Speaker 1] (3:07:40 - 3:07:41)

Yeah, I think time is more valuable.

[Speaker 2] (3:07:43 - 3:07:46)

And, you know, if the investment opportunity is good, I'd put 25 lakhs.

[Speaker 4] (3:07:46 - 3:07:51)

Okay, so Richie? How about like a lakh to 2 lakhs?

[Speaker 1] (3:07:51 - 3:08:06)

Yeah, okay. Anywhere between, I mean, 10, 15, 20, depending on the opportunity. Okay, let's say between 10 to 25 and I'll also do 25.

So, we'll add it all up and we'll have a nice, we'll find somebody really who deserves that money.

[Speaker 3] (3:08:06 - 3:08:15)

Somebody young and like brilliant. It has to be really, a lot of good hardware attempts happening. That don't get money and could deserve the money.

[Speaker 1] (3:08:16 - 3:08:24)

Done, super. So, thank you each one of you for taking how many of our 6 hours today? This is 6 hours?

[Speaker 4] (3:08:24 - 3:08:29)

It's close to. Done, done, done. Thank you so much.